User's Manual

RASPDUINO IOT SENSOR-DEVELOPMENT BOARD
MODEL: RASPDUINO

Logsun Systems

4A, Shree Sadgurunivas Society 16/5/1,

Hingne khurd, Sinhagad Road, Pune- 411051

Phone : 020 24356456 / 8956711765

Website: www.logsun.com / www.logsunonline.com

Email ID: support@logsun.com / info@logsun.com

Preface:

This manual is for the users of with in-depth details of product including reference designs and examples.

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For further details and queries contact:

Logsun Systems
4A, Shree Sadgurunivas Society 16/5/1,
Hingne khurd, Sinhagad Road, Pune- 411051
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RASPDUINO

RASPDUINO IOT SENSOR-DEVELOPMENT BOARD

CONTENTS

CHAPTER 1: INTRODUCTION

1.1 INTRODUCING RASPDUINO – THE POWERHOUSE OF INTELLIGENT IOT INNOVATION

Welcome to the future of embedded intelligence. In an era where machines speak, sense, and act with precision, the Internet of Things (IoT) is the force driving a technological revolution. IoT refers to the network of interconnected physical devices — from sensors and actuators to gateways and cloud platforms — that communicate, collect, and exchange data to create smart, responsive systems.

From smart homes and industrial automation to health monitoring and precision agriculture, IoT is not just the future — it is the now.

Amidst this rapid evolution, we proudly present **RASPDUINO** — a groundbreaking, all-in-one IoT development platform that unites **the versatility of ATmega328** with the computing power of **Raspberry Pi**. For the first time, both legendary processors are seamlessly integrated on a single board, offering **developers, researchers, and students an unmatched platform for real-world IoT innovation.**

Why RASPDUINO?

Unified Architecture: Combines high-level Linux-based processing (Raspberry Pi) with low-level hardware control (Arduino Uno).

Built for Intelligent Systems: Ideal for edge AI, smart sensing, control systems, and IoT cloud applications.

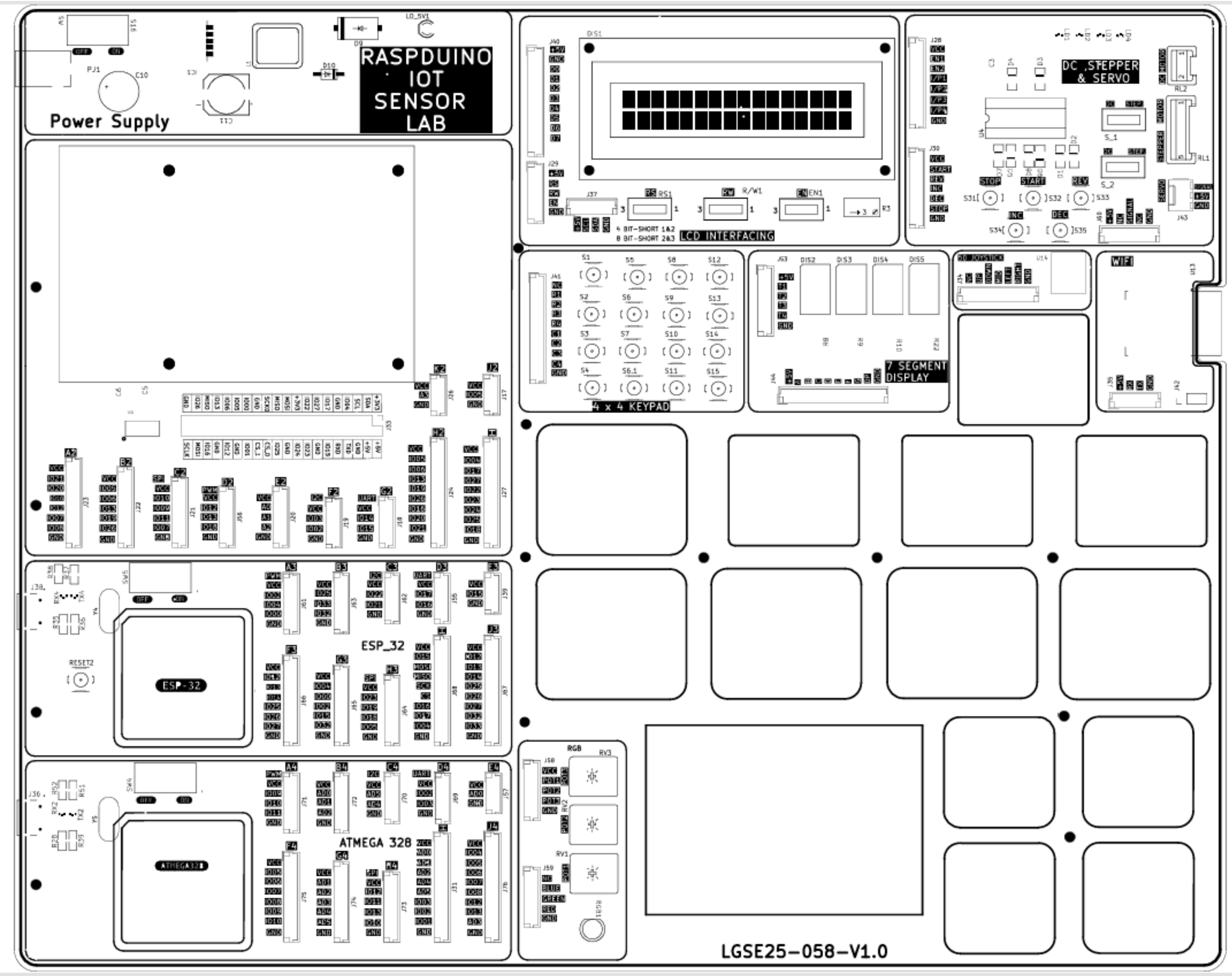
Sensor-Ready: Multiple onboard sensor interfaces ready for plug-and-play experimentation (sensor list follows).

Education + **Industry** = **RASPDUINO**: Perfect for academic training, industrial prototyping, and advanced research.

Whether you're building a smart irrigation system, an industrial IoT gateway, or a next-gen automation project — RASPDUINO is the platform that dares you to dream bigger and build smarter.

When control meets intelligence, innovation begins. Choose RASPDUINO — Your Gateway to IoT Mastery.

1.2 RASPDUINO BOARD COMPONENT LEGEND DIAGRAM



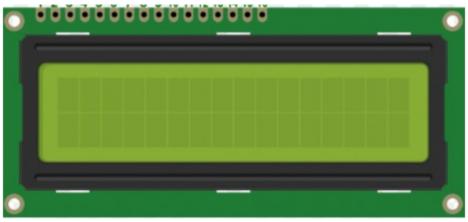
1.3 List of sensors for RASPDUINO – Highest in all trainer board category

- 16x2 LCD display
- 4x4 Keypad
- 4 Digit 7 segment display
- Stepper Motor
- Dc servo Motor
- Servo Motor
- RGB LED
- Joy stick
- Wifi Module ESP8266
- Relay
- Buzzer
- Bluetooth Module
- PIR sensor
- Ultrasonic Sensor
- Temperature humidity sensor
- LDR
- MQ135 (air quality)
- BMP 390 (Atmospheric Pressure)
- Soil moisture
- Rain Sensor
- Hall Effect
- Tilt Switch

1.4 SENSOR DESCRIPTION IN DETAIL

1. 16x2 LCD display

A 16x2 LCD display is a Liquid Crystal Display commonly used in embedded systems and electronics projects to display characters, numbers, or symbols. The term 16x2 means it can display 16 characters per line and has 2 lines, totalling 32 characters at a time.



Basic Features

- Display Capacity: 16 characters × 2 rows
- Character Format: 5x8 pixel matrix per character
- Controller: Usually based on the Hitachi HD44780 or compatible chipset
- Interface:
  - 4-bit or 8-bit parallel communication
  - Optional I2C module for easier connection

Pin Configuration (16 Pins)

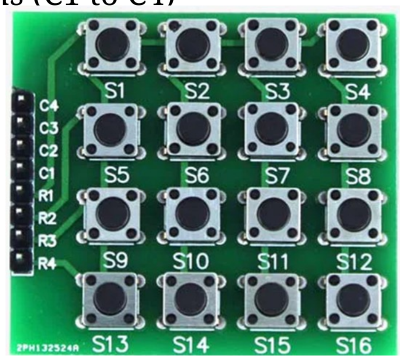
Pin No	Name	Description
1	VSS	Ground
2	VDD	+5V power supply
3	VO	Contrast adjustment (via potentiometer)
4	RS	Register Select (0 = Command, 1 = Data)
5	RW	Read/Write (0 = Write, 1 = Read)
6	E	Enable (starts data read/write)
7-14	D0–D7	Data lines (used in 4 or 8-bit mode)
15	LED+	Backlight power (+5V)
16	LED-	Backlight ground

Operating Modes

1. 8-bit Mode: Uses all 8 data lines (D0–D7)
2. 4-bit Mode: Uses only D4–D7, sends data in two 4-bit nibbles (saves GPIO pins)

2. 4 x 4 Keypad

A 4x4 keypad matrix is a type of input device made up of 16 push buttons arranged in a 4-row by 4-column grid. It is commonly used in embedded systems, microcontroller projects, and security systems for numeric or alphanumeric input. The keypad has: 4 rows (R1 to R4) and 4 columns (C1 to C4)



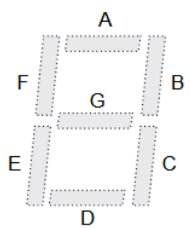
Working Principle

The keypad is essentially a matrix of switches. Pressing a key connects a row and a column, allowing current to flow and identifying which key is pressed. Microcontrollers scan the matrix by:

1. Setting rows as outputs and columns as inputs (or vice versa).
2. Pulling one row LOW at a time and checking which column reads LOW.
3. Detecting the intersection tells which key is pressed.

3. 4 Digit 7 segment display

A 4-digit 7-segment display is an electronic display device that can show four numerical digits (0–9), and sometimes a few alphabetic characters, using 7 LEDs per digit arranged in the shape of the number 8. a 4-digit 7-segment display has: 28 segments (4 digits × 7 segments), 4 decimal points (optional) and Typically 12 to 16



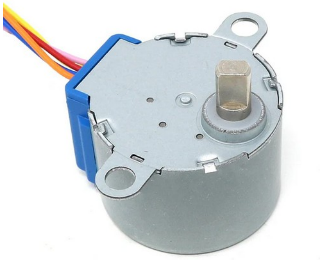
pins, depending on the type.

Types

1. **Common Cathode (CC):** All cathodes (negative) of segments are connected together per digit.
2. **Common Anode (CA):** All anodes (positive) of segments are connected together per digit.

4. **Stepper Motor**

A Stepper Motor is a type of brushless, synchronous electric motor that moves in discrete steps. It's widely used in precise position control applications, such as 3D printers, CNC machines, and robotics.



How It Works?

- The rotor (the moving part) is typically a magnet.
- The stator (stationary part) has electromagnets arranged in a circular fashion.
- When stator coils are energized in a specific sequence, the rotor aligns with the changing magnetic field.
- Each activation moves the motor by a fixed angle, called a step.

Driving Modes:

Stepper motors can be controlled in several modes:

- Full Step – Moves one full step at a time.
- Half Step – Alternates between one and two coils: smoother motion.
- Microstepping – Uses controlled current levels for very smooth and precise motion.

Control:

Stepper motors are controlled via **driver circuits**, which:

- Sequence power to coils
- Accept signals from microcontrollers (e.g., Arduino)
- Control speed and direction

5. **DC Motor**

A DC Motor (Direct Current Motor) is an electric motor that runs on direct current (DC) electricity. It converts electrical energy into mechanical rotation, making it

widely used in toys, fans, tools, robotics, and automotive systems.

Basic Working Principle

A DC motor works based on **Lorentz Force Law**:

- When current flows through a wire inside a magnetic field, it experiences a **force**.
- This force produces **torque** on the rotor (armature), causing it to rotate.

DC Motor Control

- **Speed** is controlled by voltage (more voltage = faster spin).
- **Direction** is controlled by polarity (reversing voltage reverses direction).
- Controlled using transistors**,** H-bridge drivers**, or** PWM (Pulse Width Modulation)**.**

6. Servo Motor

A Servo Motor is a special type of motor used for precise control of angular or linear position, velocity, and acceleration. It’s commonly used in robotics, RC cars, CNC machinery, and automation systems.

How a Servo Motor Works

A typical servo motor consists of:

1. **DC or Brushless motor** – Provides rotation.
2. **Gearbox** – Reduces speed and increases torque.
3. **Control Circuit** – Receives signals and adjusts motor behavior.

The servo rotates to a specific angle based on a **control signal**, usually in the form of a **PWM signal.**

Basic Working Principle

A servo motor works on the principle of **closed-loop control**:

1. A **control signal** (PWM) is sent to the servo.

2. An **internal circuit** compares this with the actual position.
3. The motor rotates to correct the error and reach the desired position.

Control Signal

- Uses **PWM** signal (typically 50 Hz or 20 ms period)
- **Pulse width** determines angle:
  - ~1 ms → 0°
  - ~1.5 ms → 90° (center)
  - ~2 ms → 180°

7. RGB LED

An RGB LED is a type of LED (Light Emitting Diode) that combines red, green, and blue LEDs into a single package. By adjusting the brightness of each of these three colors, you can create a wide range of colors, including white.

Structure of an RGB LED:

An RGB LED has **four** pins**: Red pin**, **Green pin**, **Blue pin and** c**ommon pin** (**either common cathode** or **common anode**)

There are **two types**:

- **Common Cathode (CC):** The common pin is connected to **GND (0V)**.
- **Common Anode (CA):** The common pin is connected to **VCC (positive voltage)**.

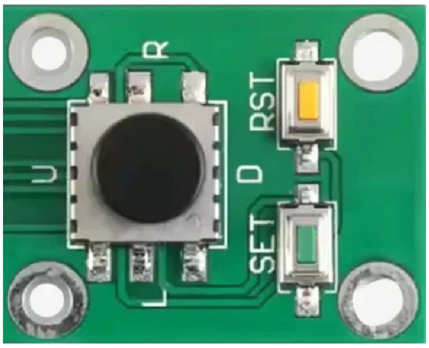
Working Principle:

Each internal LED (R, G, B) can be turned on/off or dimmed using **PWM (Pulse Width Modulation)**. Mixing different brightness levels produces different colors.

Red	Green	Blue	Output Color
255	0	0	Red
0	255	0	Green
0	0	255	Blue
255	255	0	Yellow
0	255	255	Cyan
255	0	255	Magenta
255	255	255	White
0	0	0	Off (Black/Dark)

8. 5D JOYSTICK

A 5D joystick is a versatile input device commonly used in gaming, robotics, and control systems. It provides directional control in five directions (up, down, left, right, and a center/press) and often includes additional buttons for more complex functions.



Key Features:

- **Five-Way Directional Control:** The joystick allows for movement in four cardinal directions (up, down, left, right) and a center position that can be pressed as a button.
- **Additional Buttons:** Many 5D joysticks include extra buttons (like "SET" and "RESET") that can be used for specific actions or to trigger additional commands.
- **Versatility:** These joysticks are compatible with various microcontrollers, Arduino boards, and other electronic systems.
- **Applications:** They are used in gaming controllers, navigation systems, robotics, and other projects where precise directional input is needed.

How it works?

- The joystick's lever or shaft is moved in different directions, which changes the resistance in an internal potentiometer.
- This change in resistance is interpreted as a signal by the connected microcontroller, allowing for control of various devices or programs.

Benefits:

- Precise Control: Offers precise movement and control in multiple directions.
- Versatile Functionality: Combines directional control with additional buttons for complex actions.
- Easy Integration: Designed for easy integration with various microcontroller platforms.

9. Wifi Module ESP8266

This is ESP-12E: ESP8266 Serial Port WIFI Wireless Transceiver Module For Arduino This small WIFI transceiver is the perfect solution for home Automation and IoT applications. It can be the replacement of your NRF24L01, it can talk to your WIFI router directly through the UART MCU's (Rx, Tx)

The ESP-12 module is one of the most complete of the ESP family as it allows you to use the biggest amount of pins of all of them. You can program this module to work stand alone with the Arduino IDE or with LUA as NodeMCU. The system is equipped with ESP8266 manifested leading features are: energy saving VoIP quickly switch between the sleep/wake patterns, with low-power operation adaptive radio bias, front-end signal processing functions.

Features:

1. WS2812 Led on board.
2. 802.11 b/g/n
3. Wi-Fi Direct (P2P), soft-AP
4. Integrated TCP/IP protocol stack
5. Integrated TR switch, balun, LNA, power amplifier and matching network
6. Integrated PLL, regulators, DCXO and power management units
7. +19.5dBm output power in 802.11b mode
8. Power down leakage current of <10uA
9. Integrated low power 32-bit CPU could be used as an application processor

10. SDIO 1.1 / 2.0, SPI, UART
11. STBC, 1×1 MIMO, 2×1 MIMO
12. A-MPDU & A-MSDU aggregation & 0.4ms guard interval
13. Wake up and transmit packets in < 2ms
14. Standby power consumption of < 1.0mW (DTIM3)
15. Access Point and Station Modes.

10. RELAY

Relay is an electromechanical device that uses an electric current to open or close the contacts of a switch. The single-channel relay module is much more than just a plain relay, it comprises of components that make switching and connection easier and act as indicators to show if the module is powered and if the relay is active or not.

Single-Channel Relay Module Specifications

- Supply voltage – 3.75V to 6V
- Quiescent current: 2mA
- Current when the relay is active: ~70mA
- Relay maximum contact voltage – 250VAC or 30VDC
- Relay maximum current – 10A

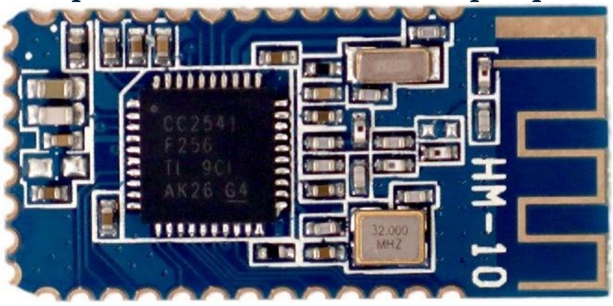
11. Buzzer

A **buzzer** is a small yet efficient component to add sound features to our project/system. It is very small and compact 2-pin structure hence can be easily used on breadboard, Perf Board and even on PCBs which makes this a widely used component in most electronic ap

There are two types of buzzers that are commonly available. The one shown here is a simple buzzer which when powered will make a Continuous Beeeeeeppp.... sound, the other type is called a readymade buzzer which will look bulkier than this and will produce a Beep. Beep. Beep. Sound due to the internal oscillating circuit present inside it. But, the one shown here is most widely used because it can be customised with help of other circuits to fit easily in our application.

12. Bluetooth Module

The HM-10 is a Bluetooth 4.0 Low Energy (BLE) module that uses the CC2541 chipset. It operates on the 2.4 GHz ISM band with GFSK modulation. The module is 26.9mm x 13mm x 2.2mm in size. It can operate in both central and peripheral roles, utilizing the FFE0 and FFE1 service UU



General:

- Bluetooth Version: Bluetooth 4.0 BLE.
- Chipset: CC2541.
- Operating Frequency: 2.4 GHz ISM band.
- Modulation: GFSK (Gaussian Frequency Shift Keying).
- Security: Authentication and encryption.
- AT Command Support: Yes, for configuration via UART.
- UART Interface: Yes, for data communication.
- Size: 26.9mm x 13mm x 2.2mm.

Power:

- Voltage: +3.3VDC.
- Current Consumption:
- Sleep Mode: 400uA~1.5mA.
- Active Mode: 8.5mA.

Features:

1. It's easy to use and completely encapsulated.
2. You can set it as a slave or master.
3. You can use the AT command set the baud rate.
4. Coverage up to 60 meters.

5. Built-in PCB antenna.
6. UART design.
7. Support bridge direct-driven mode simultaneously (no need for extra CPU)

13. PIR Sensor

PIR sensor i.e. Passive Infrared Sensor, passive word indicates PIR Sensor does not generate or radiate any energy for detection purposes.

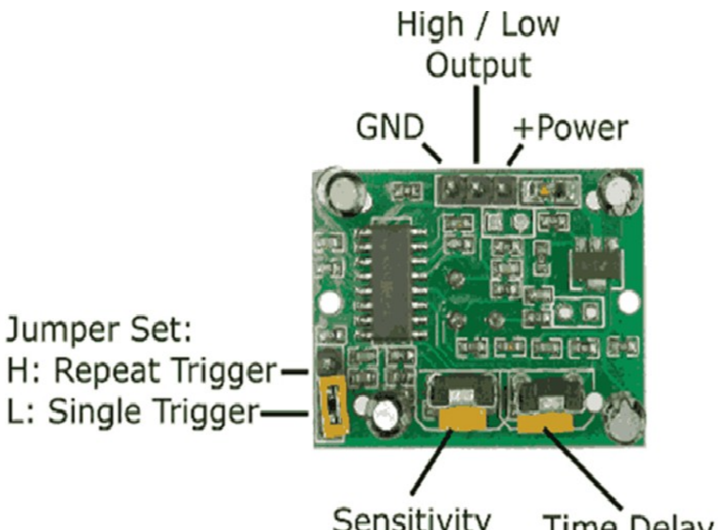
PIR Sensors don't detect or measure "**HEAT**"; they detect the infrared radiation emitted or reflected from objects.

They are small, inexpensive, low power and easy to use. They are commonly found at home, medical, factories etc. Areas.

Specification of PIR Sensor

- Detection range: up to 7 meters
- Detection angle: 110 degrees
- Operating voltage: DC 4.5V - 12V DC
- Output signal: 3.3V digital output
- Delay time: adjustable from 0.3 seconds to 5 minutes
- Operating temperature: -15°C to +70°C
- Sensitivity: Adjustable

PIR Sensor Pinout

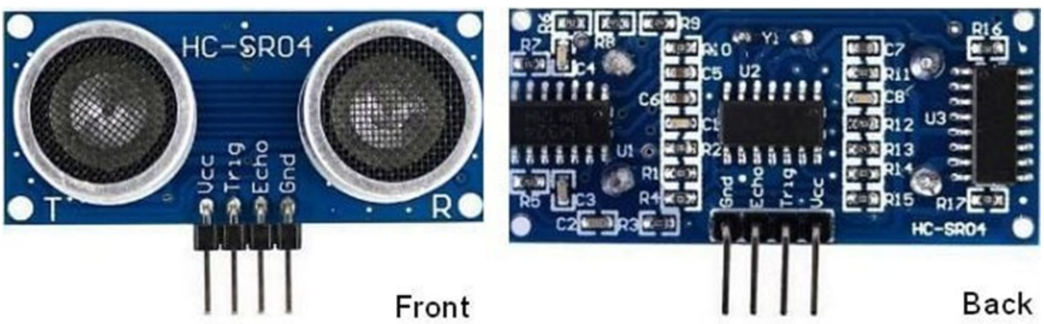


14. Ultrasonic Sensor

The ultrasonic sensor works on the principle of SONAR and RADAR system which is used to determine the distance to an object.

An ultrasonic sensor generates high-frequency sound (ultrasound) waves. When this ultrasound hits the object, it reflects as echo which is sensed by the receiver

By measuring the time required for the echo to reach to the receiver, we can calculate the distance. This is the basic working principle of Ultrasonic module to measure distance.



HC-SR04 Pin Description

- VCC: +5 V supply
- TRIG: Trigger input of sensor. Microcontroller applies 10 us trigger pulse to the HC-SR04 ultrasonic module.
- ECHO: Echo output of sensor. Microcontroller reads/monitors this pin to detect the obstacle or to find the distance.
- GND: Ground

Ultrasonic Sensor HC-SR04 Working Principle

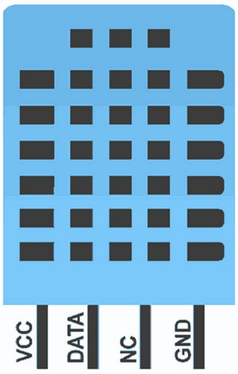
We need to transmit trigger pulse of at least 10 us to the HC-SR04 Trig Pin.

1. Then the HC-SR04 automatically sends Eight 40 kHz sound wave and wait for rising edge output at Echo pin.
2. When the rising edge capture occurs at Echo pin, start the Timer and wait for falling edge on Echo pin.
3. As soon as the falling edge is captured at the Echo pin, read the count of the Timer. This time count is the time required by the sensor to detect an object and return back from an object.

15. Temperature humidity sensor

The DHT11 is a basic, low cost digital temperature and humidity sensor.

- DHT11 is a single wire digital humidity and temperature sensor, which provides humidity and temperature values serially with one-wire protocol.
- DHT11 sensor provides relative humidity value in percentage (20 to 90% RH) and temperature values in degree Celsius (0 to 50 °C).
- DHT11 sensor uses resistive humidity measurement component, and NTC temperature measurement compo



DHT11 Pinout:

- DHT11 is a 4-pin sensor, these pins are VCC, DATA, GND and one pin is not use.

Pin Description:

Pin No.	Pin Name	Pin Description
1	VCC	Power supply 3.3 to 5.5 Volt DC
2	DATA	Digital output pin
3	NC	Not in use

4	GND	Ground

Specification of the DHT11 sensor

- Power supply: 3.3 to 5V DC
- Current consumption: max 2.5mA
- Operating range: 20-80% RH, 0-50°C
- Humidity measurement range: 20-90% RH
- Humidity measurement accuracy: ±5% RH
- Temperature measurement range: 0-50°C
- Temperature measurement accuracy: ±2°C
- Response time: 1s
- Sampling rate: 1Hz (1 sample per second)
- Data output format: single-bus digital signal
- Data transmission distance: 20-30m (at open air)

16. LDR Sensor

LDR (Light Dependent Resistor) as the name states is a special type of resistor that works on the photoconductivity principle means that resistance changes according to the intensity of light. Its resistance decreases with an increase in the intensity of light. It is often used as a light sensor, light meter, Automatic street light, and in areas where we need to have light sensitivity. LDR is also known as a Light Sensor. LDR are usually available in 5mm, 8mm, 12mm, and 25mm dimensions.

LDR Working Principle

It works on the principle of photoconductivity whenever the light falls on its photoconductive material, it absorbs its energy and the electrons of that photoconductive material in the valence band get excited and go to the conduction band and thus increasing the conductivity as per the increase in light intensity.

Also, the energy in incident light should be greater than the bandgap gap energy so that the electrons from the valence band got excited and go to the conduction band.

We have a detailed article on working, circuit, and construction of LDR.

The LDR has the highest resistance in dark around 1012 Ohm and this resistance decreases with the increase in Light.

17. MQ135 (Air Quality)

A device that is used to detect or measure or monitor the gases like ammonia, benzene, sulfur, carbon dioxide, smoke, and other harmful gases are called as an air quality gas sensor. The MQ135 air quality sensor, which belongs to the series of MQ gas sensors, is widely used to detect harmful gases, and smoke in the fresh air. This article gives a brief description of how to measure and detect gases by using an MQ135 air quality sensor.

What is an MQ135 Air Quality Sensor?

An MQ135 air quality sensor is one type of MQ gas sensor used to detect, measure, and monitor a wide range of gases present in air like ammonia, alcohol, benzene, smoke, carbon dioxide, etc. It operates at a 5V supply with 150mA consumption. Preheating of 20 seconds is required before the operation, to obtain the accurate output.

It is a semiconductor air quality check sensor suitable for monitoring applications of air quality. It is highly sensitive to NH3, NOx, CO2, benzene, smoke, and other dangerous gases in the atmosphere. It is available at a low cost for harmful gas detection and monitoring applications.

If the concentration of gases exceeds the threshold limit in the air, then the digital output pin goes high. The threshold value can be varied by using the potentiometer of the sensor. The analog output voltage is obtained from the analog pin of the sensor, which gives the approximate value of the gas level present in the air.

Specifications and Features:

The **MQ135 air quality sensor specifications and features** are listed below.

- It has a wide detection scope.
- High sensitivity and faster response.
- Long life and stability.
- The operating voltage: +5V.
- Measures and detects NH_3, alcohol, NO_x, Benzene, CO_2, smoke etc.
- Range of analog output voltage: 0V-5V.
- Range of digital output voltage: 0V-5V (TTL logic).
- Duration of preheating: 20 seconds.
- Used as an analog or digital sensor.
- The potentiometer is used to vary the sensitivity of the digital pin.
- Heating Voltage: 5V±0.1.
- Load resistance is adjustable.

18. BMP 390 (Atmospheric Pressure)

The BMP390 is a very small, low-power and low-noise 24-bit absolute barometric pressure sensor. The digital, high-performance sensor is ideally suited for a wide range of altitude tracking applications. This new BMP390 pressure sensor offers outstanding design flexibility, providing a single package solution that customers can easily integrate into a multitude of existing and upcoming devices such as smartphones, GPS modules, wearables, hearables and drones.

Features

- Onboard BMP390 sensor for measuring barometric pressure, altitude and temperature
- Supports I2C communication, I2C address configurable, with I2C bus cascading support
- Onboard voltage translator, compatible with 3.3V/5V level

19. Soil moisture

This **soil moisture sensor module** is used to detect the moisture of the soil. It measures the volumetric content of water inside the soil and gives us the moisture level as output. The module has both digital and analog outputs and a potentiometer to adjust the threshold level.

Soil Moisture Sensor Module Pinout Configuration

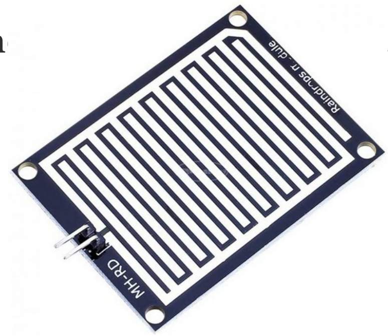
Pin Name	Description
VCC	The Vcc pin powers the module, typically with +5V
GND	Power Supply Ground
DO	Digital Out Pin for Digital Output.
AO	Analog Out Pin for Analog Output

Soil Moisture Sensor Module Features & Specifications

- Operating Voltage: 3.3V to 5V DC
- Operating Current: 15mA
- Output Digital - 0V to 5V, Adjustable trigger level from preset
- Output Analog - 0V to 5V based on infrared radiation from fire flame falling on the sensor
- LEDs indicating output and power
- PCB Size: 3.2cm x 1.4cm
- LM393 based design
- Easy to use with Microcontrollers or even with normal Digital/Analog IC
- Small, cheap and easily available

20. Rain Sensor

Raindrop Sensor is a tool used for sensing rain. It consists of two modules, a **rain board** that detects the rain and a **control module**, which compares the analog value, and converts it to a digital value. The raindrop sensors can be used in the automobile sector to control the windshield wipers automatically, in the agriculture sector to sense rain and it is also used in h .



Pin Configuration of Rain Sensor:

Sr. No:	Name	Function
1	VCC	Connects supply voltage- 5V
2	GND	Connected to ground
3	D0	Digital pin to get digital output
4	A0	Analog pin to get analog output

Raindrop Sensor Features:

- Working voltage 5V
- Output format: Digital switching output (0 and 1), and analog voltage output AO
- Potentiometer adjust the sensitivity
- Uses a wide voltage LM393 comparator
- Comparator output signal clean waveform is good, driving ability, over 15mA
- Anti-oxidation, anti-conductivity, with long use time

- With bolt holes for easy installation

21. Hall Effect

Hall effect sensors are electronic components that detect the presence and strength of a magnetic field. Specifications vary depending on the specific sensor type and application, but common features include digital or analog output, operating voltage, sensitivity, and temperature range. Some sensors also have reverse polarity protection and can be used for both pole detection.

Key Specifications:

- Output Type:
- Digital: Outputs a high or low signal, indicating if a magnet is present.
- Analog: Outputs a voltage that varies proportionally to the magnetic field strength.
- Operating Voltage: The voltage required for the sensor to function.
- Sensitivity: The change in output voltage or current per unit change in magnetic field strength (typically measured in Gauss or Tesla).
- Operating Temperature Range: The temperature range over which the sensor can operate reliably.
- Output Current: The maximum current the sensor can provide.
- Power Consumption: The amount of current the sensor draws from the power supply.

- Magnetic Field Strength: The range of magnetic field strengths the sensor can detect.
- Sensitivity Drift: The change in sensitivity over temperature.

Applications:

- Position sensing: Detecting the position of rotating or moving parts.
- Speed sensing: Measuring the speed of rotating objects.
- Motor control: Providing feedback for motor control systems.
- Current sensing: Measuring the current flowing through conductors.
- Magnetic field measurement: Measuring the strength of magnetic fields.

22. Tilt Switch

Tilt switches are simple electromechanical sensors that open or close an electrical circuit when tilted. They are typically used to detect changes in orientation and can be found in various applications, including alarms, motion detection, and level sensing. Tilt switches have a variety of specifications, including operating voltage, switching angle, contact resistance, ance.

Basic Functionality:

- **Orientation Sensing:**
  Tilt switches respond to changes in their physical orientation, typically by sensing when the switch is tilted beyond a certain angle.
- **Electrical Contacts:**
  They have two or more electrical contacts that are either in contact or

separated based on the tilt angle.

Common Specifications:

- **Operating Voltage:**
  Typically, tilt switches can operate with a range of voltages, often between 3.3V and 5V, or even higher.
- **Switching Angle:**
  This refers to the angle at which the switch transitions between its open and closed states.
- **Contact Resistance:**
  This is a measure of the resistance between the switch's contacts when they are closed, and it's typically low for good electrical conductivity.
- **Insulation Resistance:**
  This measures the resistance between the switch's contacts when they are open, and it's typically high for good isolation.
- **Sensitivity:**
  How much tilt is required to trigger the switch, measured in degrees or other angular units.
- **Lifespan:**
  The number of cycles the switch can be switched on and off before degradation.
- **Dimensions:**
  The physical size of the switch, including its length, diameter, and height.
- **Environmental Ratings:**
  Some tilt switches are designed for specific environments, such as those with high temperatures, moisture, or pressure.

Applications:

- Alarms: Triggering alarms when an object is tilted or tipped.
- Motion Detection: Detecting movement by sensing changes in tilt.
- Level Sensing: Monitoring the level of a container or tank.
- Robotics: Detecting the orientation of a robot.
- Gaming: Used in game controllers to detect player input.

CHAPTER 2: INTRODUCTION TO RASPBERRY PI

The **Raspberry Pi** is single-board computer developed by the Raspberry Pi Foundation to promote computer science education and experimentation. Originally designed for educational purposes, it has evolved into a powerful tool for hobbyists, researchers, educators, and developers across domains like IoT, AI, robotics, automation, and embedded systems.

The Raspberry Pi has revolutionized the landscape of embedded computing by democratizing access to powerful and flexible computing platforms. Its integration in the **RASPDUINO** hardware marks a significant step toward enabling the development of next-generation intelligent electronic systems, where both **computational power and hardware-level control** are available on the same board.

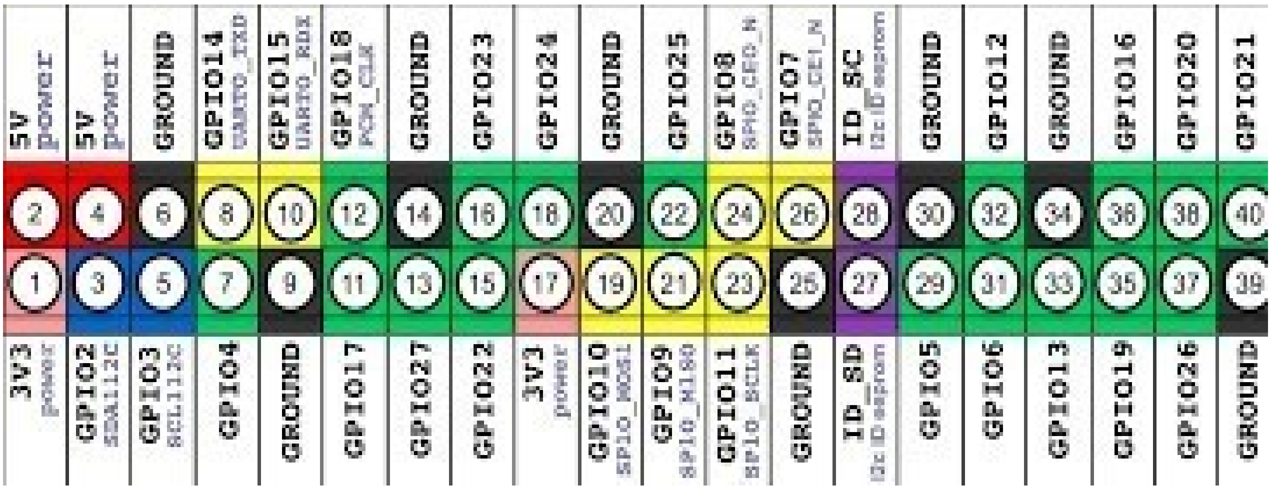
Raspberry Pi-4 Model-B

Specification Details:

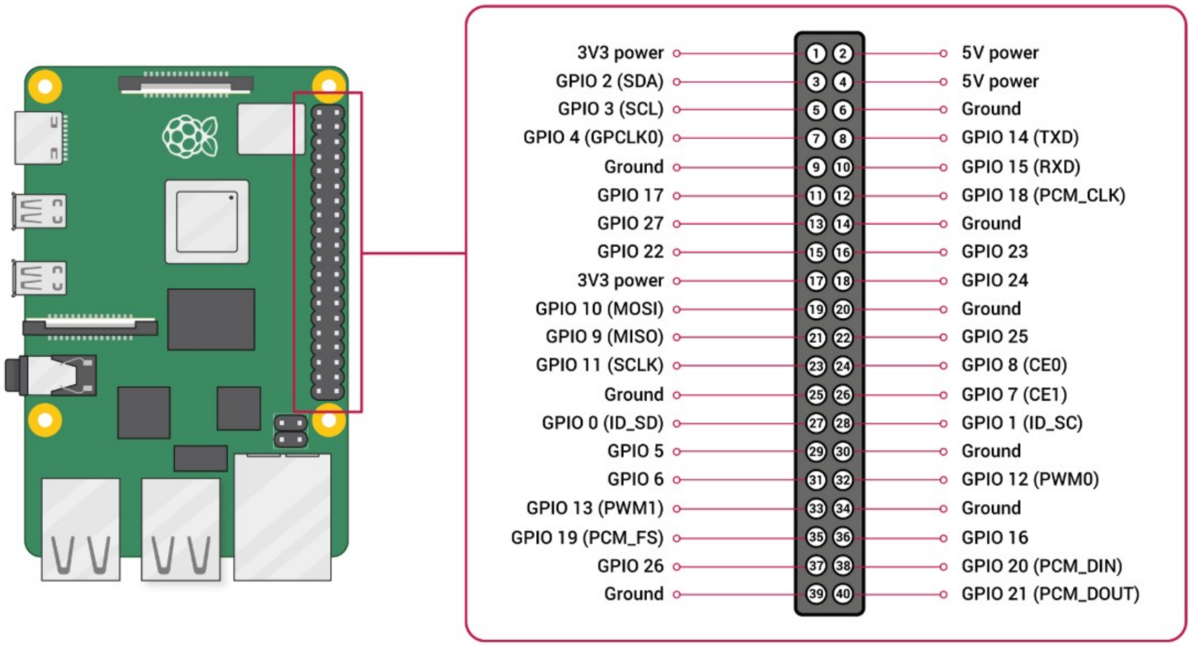
- ✓ **Processor**
    - Broadcom BCM2711
    - Quad-core Cortex-A72 (ARM v8) 64-bit SoC
    - Clock Speed: 1.5 GHz
- ✓ **Memory (RAM) Options**
    - 4 GB LPDDR4-3200 SDRAM variants
- ✓ **Connectivity**
    - Wi-Fi: 802.11 b/g/n/ac (dual-band 2.4 GHz and 5 GHz)
    - Bluetooth: 5.0, BLE
    - Gigabit Ethernet port
    - 2 × USB 3.0 ports
    - 2 × USB 2.0 ports
    - 40-pin GPIO header (backward-compatible)
- ✓ **Video & Display**
    - 2 × micro-HDMI ports, up to 4K resolution:
    - 1 × 4Kp60 or 2 × 4Kp30 output support
    - H.265 (4Kp60 decode), H.264 (1080p60 decode, 1080p30 encode)

- ✓ **Storage**
  - microSD card slot for operating system and data storage
  - Booting support from USB as well
- ✓ **Power Supply**
  - 5V / 3A DC via USB-C connector
  - Power over Ethernet (PoE) support via separate HAT
  - 5V DC via GPIO header
- ✓ **Camera & Display Interfaces**
  - CSI camera port (for Raspberry Pi camera module)
  - DSI display port (for official touch display)
- ✓ **Audio**
  - 4-pole stereo audio and composite video port
  - HDMI audio output
- ✓ **Dimensions & Weight**
  - Size: 85.6 mm × 56.5 mm
  - Weight: Approximately 46 grams
- ✓ **Operating System**
  - Raspberry Pi OS (Linux-based)
  - Also supports Ubuntu, Windows IoT Core, and third-party Linux distro
- ✓ **Environment:**
  - Operating temperature, 0–50°C
- ✓ **Compliance:**
  - For a full list of local and regional product approvals, please visit www.raspberrypi.org/products/raspberry - pi-4-model-b.

RASPBERRY PI GPIO DETAILS:



Overview of Raspberry Pi 40-Pin GPIO Header:



Overview of Raspberry Pi 40-Pin GPIO Header

✓ **Power Pins**

- 3.3 V Power: pins 1 & 17
- 5 V Power: pins 2 & 4
- Ground (GND): pins 6, 9, 14, 20, 25, 30, 34, 39

✓ **I²C Interface**

- SDA (GPIO 2): physical pin 3
- SCL (GPIO 3): physical pin 5

✓ **SPI Interface**

- MOSI (GPIO 10): pin 19
- MISO (GPIO 9): pin 21
- SCLK (GPIO 11): pin 23
- CE0 (GPIO 8): pin 24
- CE1 (GPIO 7): pin 26

✓ **UART Serial**

- TXD (GPIO 14): pin 8
- RXD (GPIO 15): pin 10

✓ **PWM (Hardware)**

- GPIO 12: pin 32
- GPIO 13: pin 33
- GPIO 18: pin 12
- GPIO 19: pin 35

✓ **PCM (I²S Audio)**

- GPIO 18 (CLK): pin 12
- GPIO 20 (DIN): pin 38
- GPIO 21 (DOUT): pin 40
- GPIO 19 (FS): pin 35

✓ **Reserved / EEPROM**

- GPIO 0 & GPIO 1 (pins 27 & 28) reserved for HAT EEPROM and should remain unused for other purpose.

3.1 HOW TO INSTALL OPERATING SYSTEM IN RASPBERRY PI SD CARD?

This Raspberry Pi is the best way to get started on working with the IoT (Internet of Things) and to build your knowledge to expand it to other applications. So, we made a step-by-step guide for Pi beginners so you can learn various methods that can be used to make your application of choice.

We have divided the whole Raspberry Pi IoT project into two parts. The first part consists of setting up the Raspberry Pi and interfacing it with the sensors & actuators. The second part covers building the various applications and connecting it to the server.

If you aren't familiar with Linux and terminal, check out Basic Linux Commands for Beginners.

Basic Commands -

1. **ls** — Use the **"Is"** command to know what files are in the directory you are in. You can see all the hidden files by using the command **“ls”.**

```
nayso@Alok-Aspire:~$ ls
Desktop          itsuserguide.desktop  reset-settings     VCD_Copy
Documents        Music                 School_Resources   Videos
Downloads        Pictures              Students_Works_10
examples.desktop Public                Templates
GplatesProject   Qgis Projects         TuxPaint-Pictures
```

2. **cd** — Use the **"cd"** command to go to a directory. Remember, this command is case sensitive, and you have to type in the name of the folder exactly as it is. But there is a problem with these commands. Imagine you have a folder named “Raspberry Pi”. In this case, when you type in **“cd Raspberry Pi”**, the shell will take the second argument of the command as a different one, so you will get an error saying that the directory does not exist. Here, you can use a backward slash. That is, you can use **“cd Raspberry\ Pi”** in this case. Spaces are denoted like this: If you just type **“cd”** and press enter, it takes you to the

home directory. To go back from a folder to the folder before that, you can type “**cd** ..” . The two dots represent back.

```
nayso@Alok-Aspire:~$ cd Downloads
nayso@Alok-Aspire:~/Downloads$ cd
nayso@Alok-Aspire:~$ cd Raspberry\ Pi
nayso@Alok-Aspire:~/Raspberry Pi$ cd ..
nayso@Alok-Aspire:~$
```

1. **echo** — The "**echo**" command helps us move some data, usually text into a file.
2. **cat** — Use the **cat** command to display the contents of a file. It is usually used to easily view programs.

```
nayso@Alok-Aspire:~/Desktop$ echo hello, my name is alok >> new.txt
nayso@Alok-Aspire:~/Desktop$ cat new.txt
hello, my name is alok
nayso@Alok-Aspire:~/Desktop$ echo this is another line >> new.txt
nayso@Alok-Aspire:~/Desktop$ cat new.txt
hello, my name is alok
this is another line
```

3. **nano, vi, jed** — **nano** and **vi** are already installed text editors in the Linux command line. The **nano** command is a good text editor that denotes keywords with color and can recognize most languages. And **vi** is simpler than **nano**. You can create a new file or modify a file using this editor. You can save your files after editing by using the sequence Ctrl+X, then Y (or N for no).

4. **sudo** — A widely used command in the Linux command line, **sudo** stands for "SuperUser Do". So, if you want any command to be done with administrative or root privileges, you can use the **sudo** command.

Note: We’ll be remotely accessing the Raspberry Pi’s terminal through SSH, so you won’t need a dedicated monitor, mouse or keyboard in any part of this tutorial. You would also want to download the following software:

- Win32 Disk Imager
- PuTTY
- Advanced IP Scanner

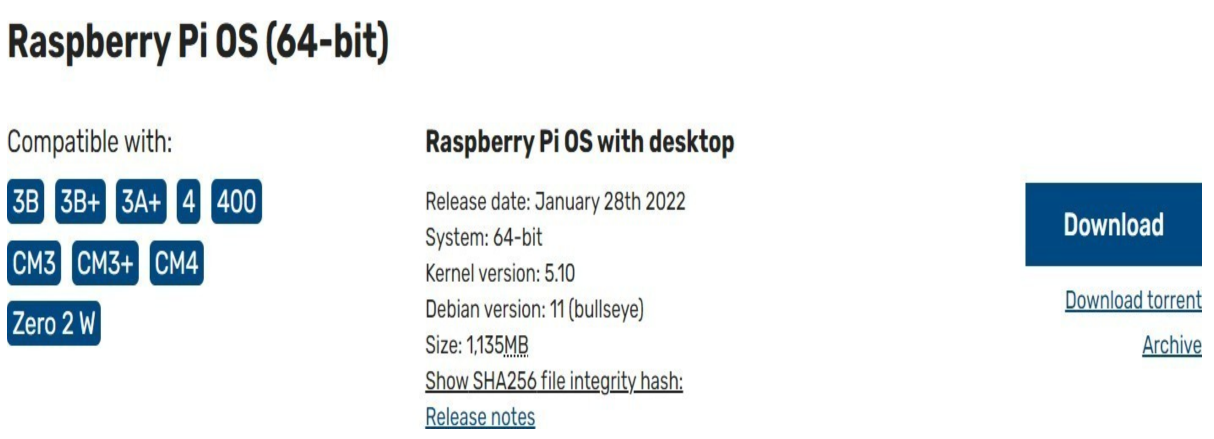
Tips & Tricks:

- You can use the **clear** command to clear the terminal if it gets filled up with too many commands.
- **TAB** can be used to fill up in terminal. For example, you just need to type "**cd Doc**" and then **TAB** and the terminal fills the rest up and makes it "**cd Documents**".
- **Ctrl+C** can be used to stop any command in terminal safely. If it doesn't stop with that, then **Ctrl+Z** can be used to force stop it.
- You can exit from the terminal by using the **exit** command.
- You can power off or reboot the computer by using the command **sudo halt** and **sudo reboot**.

3.2 SETTING UP THE PI FOR THE RASPBERRY PI IOT PROJECT

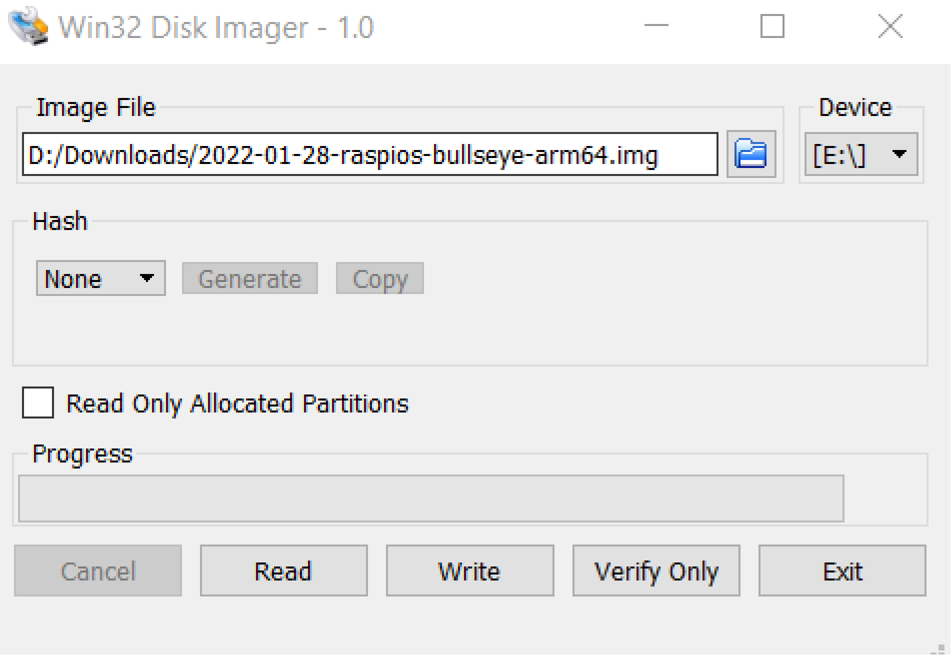
If you already have a Raspberry Pi set up, move over to the next step to start with your Raspberry Pi IoT project. Otherwise, download the Raspbian OS for your Pi. There are many other distributions you can use, but Raspbian remains the most common and convenient for beginners. Visit https://www.raspberrypi.org/downloads/raspbian/ to download the Raspbian OS.

Extract the ***.img** file from the downloaded zip folder and write it to your SD card.

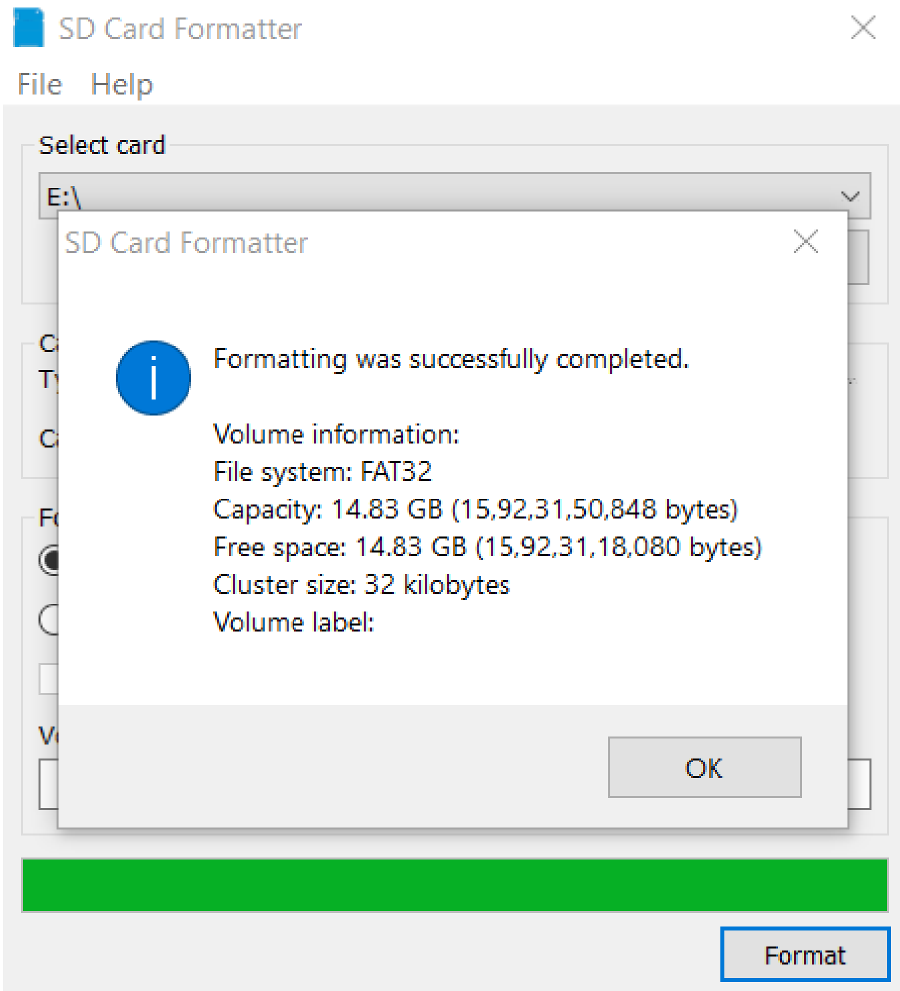


For Windows Users:

1. Insert the SD card reader with the SD card in it. Figure out the Drive assigned to it.
2. Run Win32 Disk Imager (you may have to run this as administrator), and select the extracted image file and the drive letter. Be very careful to select the correct drive—you do not want to unintentionally destroy other data. It should look something like this:



3. Click "Write". Wait for the process to complete and eject the SD card.



3.3 HOW TO CONNECT RASPBERRY PI ON WINDOWS PLATFORM?

Booting Your Raspberry Pi IoT System:

Getting a dedicated monitor, mouse, and keyboard to use your Raspberry Pi might become an unnecessary hassle. Access to the terminal is sufficient to get most things done. So we eliminate the need for extra hardware by logging into the Pi using your personal laptop through SSH. Latest versions of Raspbian come with SSH enabled by default, so you can run Pi remotely even while setting it up for the first time.

3.4 SETTINGS FOR RASPBERRY PI: AFTER USING IT FIRST TIME

(You can skip this step, as we have already done this step)

1. Select location, language and time location.
2. Click on US keyboard and English Language.
3. Set a new password and don't keep it as "raspberry" as that's the default password and it won't change.
4. If you have internet access select your network and enter your password to connect.
5. Let the system check for software updates and install those updates*.
6. Click on Finish
7. Click on the Pi Menu > Preferences click on 'Raspberry Pi configuration'
8. Go to interfaces.
    a. All the interfaces are disabled by default.
    b. Enable all the interfaces except for Remote GPIO (Enable if you really need it).
    c. Click OK and it will ask you to reboot.
    d. Click on reboot.

***If the software update fails, then open terminal (Ctrl + Alt + T) Type "sudo apt-get update && sudo apt-get upgrade"**

Press 'Y' and press enter if asked.

'update' will update the repositories and 'upgrade' will install all the new updates. Reboot once the process is complete.

Complete step (8. b), before reboot to avoid rebooting too many times.

3.5 RASPBERRY PI 4 LIBRARIES/PACKAGES INSTALLATION

1) Install AdaFruit Library for DHT22 Practical it is required.

1. Open Terminal (Ctrl + Alt + T).
2. Type "sudo apt-get update && sudo apt-get upgrade"
3. Type "sudo apt-get install python3-dev python3-pip"

4. Type “sudo python3 -m pip install --upgrade pip setup tools wheel”
5. Type “sudo pip3 install Adafruit_DHT”

Don’t forget to add ‘sudo’ as it gives you root access.

2) Enable SSH for MQTT

1. Type “sudo apt-get install python3-seria**l”**
2. Copy “dmesg | grep tty” and paste.
    1. This will show you all the available ports.
    2. ttyAMA0 might be the only enabled port.
3. **Type “sudo rasp-config”**
    1. Use arrow pointer to select the correct option, then press tab to select **Yes** or **No** option.
    2. Select Interfacing Options
    3. Select P6 Serial
        a. Select ***NO*** for “Login shell access over serial”
        b. Select ***YES*** for “Enable serial port hardware”
        c. Select ***YES*** to Reboot now.

3) pcf8575 libraries installation “pip install pcf

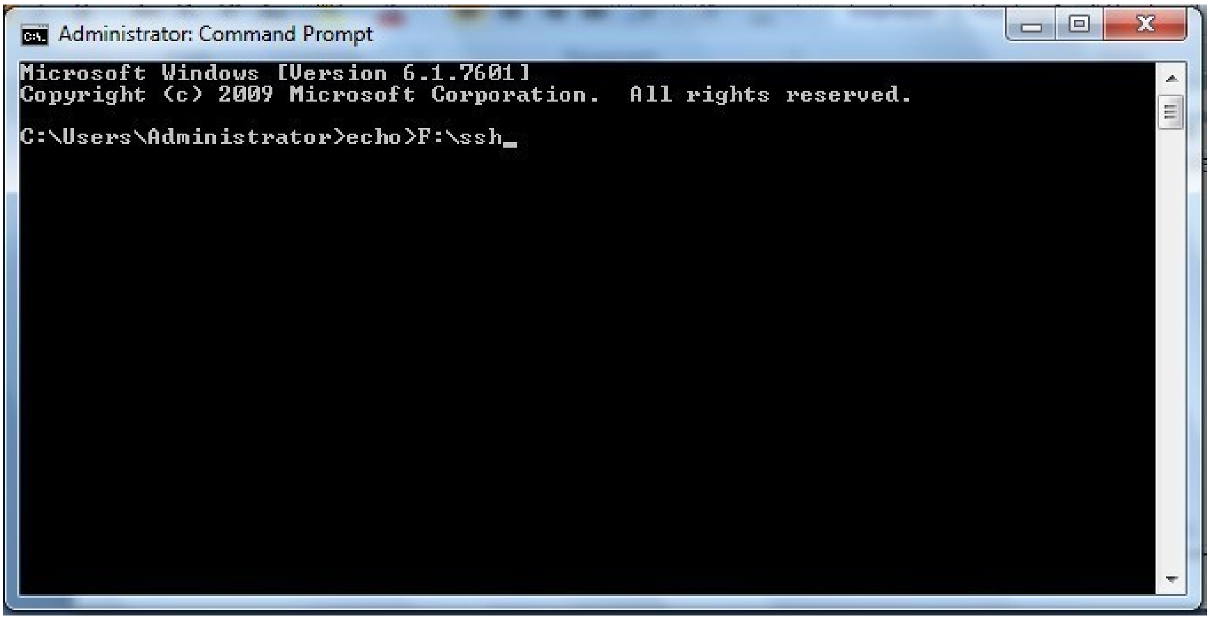
NOTE:

If above libraries didn’t work/or you need to use other libraries than this then go to [PyPI ·]The Python Package Index and use that library to copy paste content in terminal window of raspberry pi the short cut is “ctrk+shift+v”

If SSH is not enabled then do the following procedure:

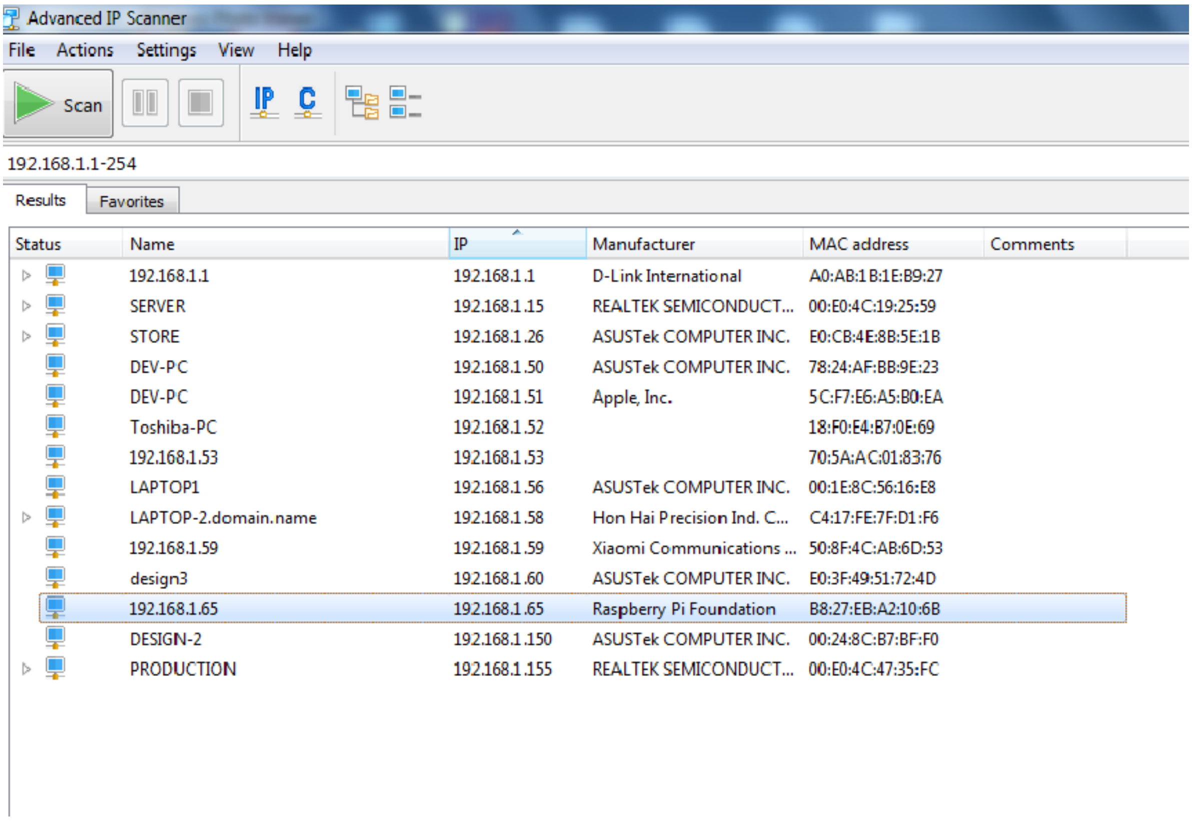
- Click on Start - All Programs – Accessories - Command Prompt
- Enter command : *echo>F:\ssh<Enter>*

You will see the window like this:

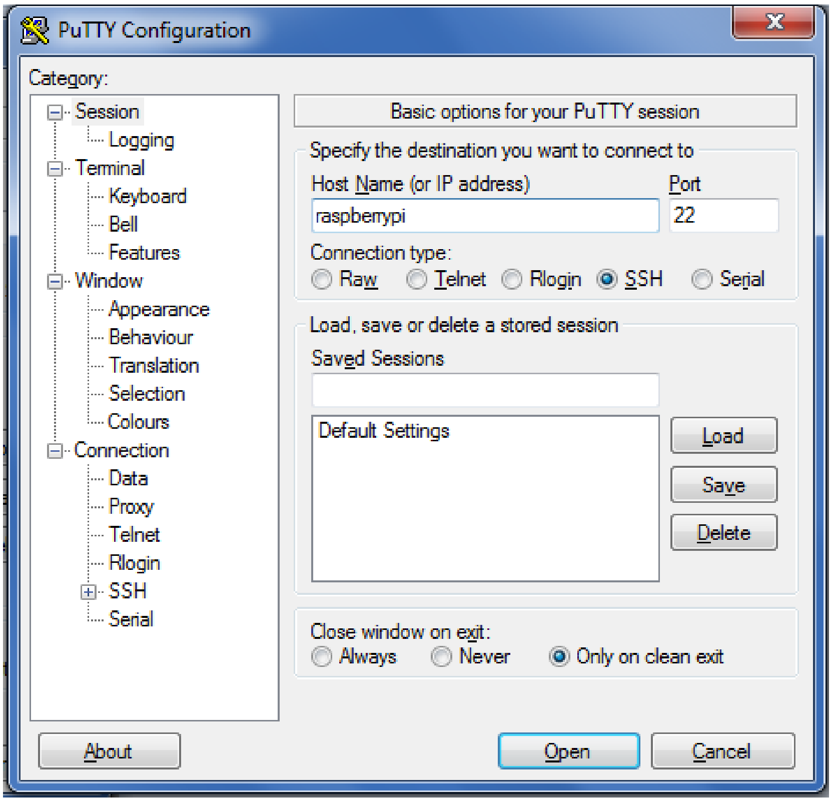


Note: *Here F is Removable Disk Drive Letter like of SD Card. Ensure your SD Card drive letter & make changes in above command accordingly.*

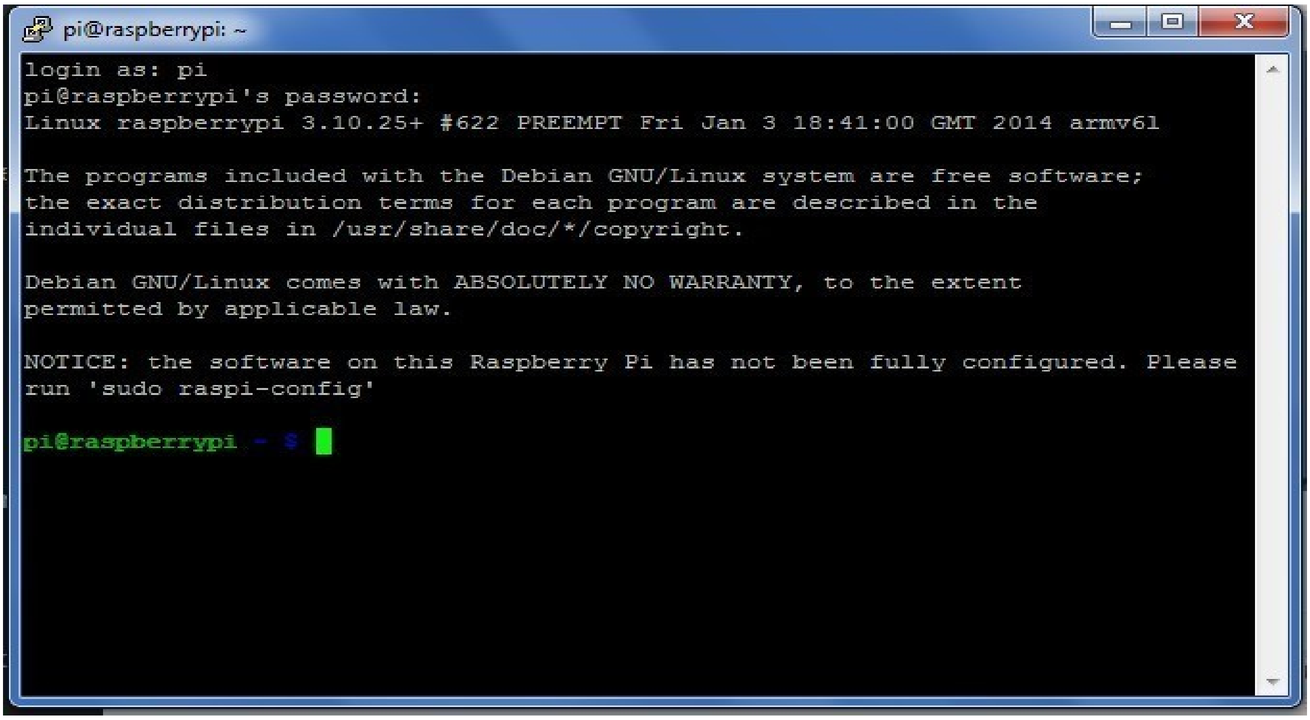
- **Get a router with DHCP enabled**. This is necessary because we want our Pi to have a unique IP address to be able to connect to it. Connect both your laptop and your Pi to the router via Ethernet cables. Your laptop and Pi now share a local area network and can identify each other using their unique IP address.
- To use SSH, you'll need the **IP address of your Pi**.
- Connect Raspberry pi with 5V power supply
- Open the Advance IP Scanner Software
- Run the Advanced IP Scanner. This will list out the IP addresses of all devices on your network and their manufacturer. You'll see something like this:



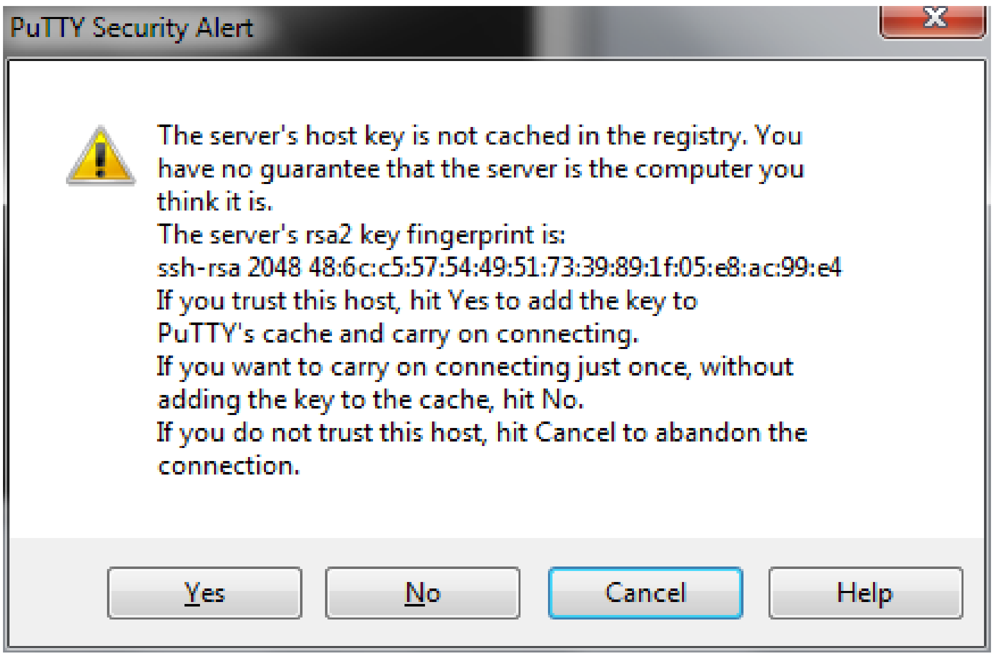
- Use PuTTY to access Pi's Terminal. Run PuTTY and simply enter the IP address determined in the above step.\



Subsequently, you will receive a login prompt. Use login id as **pi** and password as **raspberry**.



Side Note: For first-time login, you'll receive a warning for a security alert. Click on "Yes" and proceed. You have now opened a terminal session on your Raspberry Pi, which can be accessed through your Windows laptop.



3.6 HOW TO CONNECT RASPBERRY PI ON LINUX PLATFORM?

- To know your Raspberry Pi IP address, turn ON your laptop/desktop Wi-Fi.
- Connect Ethernet cable between Raspberry Pi & laptop/desktop.
- Click on Wi-Fi icon.
- Click on VPN connection – configure VPN
- Click on add
- Choose connection type as Ethernet
- Click on create
- Connection name- Raspberry Pi (You can assign any name)
- Go to IP4 settings tab.
- Click on Method & select as share to other computers.
- Click on save.
- Yu will see Raspberry Pi Connected notification.
- Open your terminal window
- enter the following command: cat /var/lib/misc/dnsmasq.leases<Enter>
- You will get Raspberry Pi IP address.
- Now enter the following command : ssh pi@10.42.0.66 <Enter>
- Enter the password as raspberry.
- You'll receive a warning for a security alert. Click on "Yes" and proceed. You have now opened a terminal session on your Raspberry Pi, which can be accessed through your Windows laptop.
- Please note that the cursor won't move forward while entering the password due to default settings.

CHAPTER 4: EXPERIMENTS ON RASPBERRY PI

EXPERIMENT NO. 1: LCD INTERFACE

AIM: - To study the interfacing of LCD (16*2) with the Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 8 pin connector, 5v,3A power supply, LCD.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 8 pin connector from the board to "LCD" section.
5. Open the saved folder on PI4 with name Raspberry Pi sample codes
6. Now we are into the folder **Raspberry Pi sample codes**.
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**
8. To run this specific program enter the command >> **python lcd.py**
9. If you want to see program enter the command >> **sudo nano lcd.py**
10. To exit from the program press **Ctrl-Z.**
11. You will observe that it displays "LOGSUN SYSTEMS" on LCD display.

Note: First connect relimate connector then run the program.

EXPERIMENT NO. 2: I2C LCD DISPLAY

AIM: - To study the interfacing of I2C LCD (16*2) with the Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin connector, 5v,3A power supply, LCD.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "LCD" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   a. *cd /home/pi/Raspberry Pi sample codes .*
6. Now we are into the folder
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**
8. To run this specific program enter the command >> **python lcd_pcf8574.py**
9. If you want to see program enter the command >> **sudo nano lcd_pcf8574.py**
10. To exit from the program press **Ctrl-Z.**
11. You will observe that it displays "**LOGSUN SYSTEMS, I2C DEMO**" on LCD display.

EXPERIMENT NO. 3: LCD WITH KEYPAD INTERFACING

AIM: To study the interfacing of LCD and Keypad with Raspberry Pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, two 8 pin connectors, 5v, 3A power supply, LCD, 4*4 matrix keypad.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. Connect one 8-pin connector from the board to "KEYPAD" section and another 8-pin connector to "LCD" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   a. *cd /home/pi/Raspberry Pi sample codes .*

6. Now we are into the folder**Raspberry Pi sample codes** .
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**
8. To run this specific program enter the command >> **python keypad.py**
9. If you want to see program enter the command >> **sudo nano keypad.py**
10. To exit from the program press **Ctrl-Z.**
11. You will observe that when you press a key (0-F) it will show on LCD display.

EXPERIMENT NO. 4: STEPPER MOTOR WITH RASPBERRY PI

AIM: To study the interfacing of stepper motor with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 8 pin connector, 5v, 3A power supply, stepper motor, external12v dc power supply.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 8 pin connector from the board to "STEPPER MOTOR" section; also connect the 12V power supply externally.
5. Open the Terminal window of Raspberry pi and enter the command >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes**
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**
8. To run this specific program enter the command >> **python stepper.py**
9. If you want to see program enter the command >> **sudo nano stepper.py**
10. To exit from the program press **Ctrl-Z**.
11. You will observe that when you run the program, the motor will automatically start rotating in forward direction.

EXPERIMENT NO. 5: SERVO MOTOR WITH RASPBERRY PI

AIM: To study interfacing of servo motor with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin connector, 5v, 3A power supply.

INTRODUCTION: - Servo Motor is nothing but a simple electrical motor where rotation of the motor is required for just a certain angle not continuously for long period of time.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "SERVO" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes**
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**
8. To run this specific program enter the command >> **python servo.py**
9. If you want to see program enter the command >> **sudo nano servo.py**
10. To exit from the program press **Ctrl-Z.**
11. When we run the program the servo motor will start rotating in clockwise direction and after some time interval it will start rotating in Anti-clockwise direction.

EXPERIMENT NO. 6: RELAY-BUZZER WITH RASPBERRY PI

AIM: To study interfacing of relay-buzzer with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin connector, 5v,3A power supply.

INTRODUCTION: - A relay is an electrically operated switch. Relays are used where it is necessary to control a circuit by a separate low-power signal, or where several circuits must be controlled by one signal. Buzzer is an audio signaling device which may be mechanical or electromechanical.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "RELAY-BUZZER" section.
5. Open the Terminal window of Raspberry pi and enter the command >> >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes**. To see the list of files into SENSOR BOARD folder enter command >> **ls**
7. To run this specific program enter the command >> **python relay_buzzer.py**
8. If you want to see program enter the command >> **sudo nano relay_buzzer.py**
9. To exit from the program press **Ctrl-Z.**
10. You will observe that relay is ON for 5sec and OFF for 5 sec and buzzer will also ON at a time.

EXPERIMENT NO. 7: ULTRASONIC SENSOR WITH RASPBERRY PI

AIM: To study interfacing of Ultrasonic Sensor with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin connector, 5v,3A power supply.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "ULTRASONIC SENSOR" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes**
7. To see the list of files into SENSOR BOARD folder enter command >> **ls**
8. To run this specific program enter the command >> **ultralcd.py**If you want to see program enter the command >> **sudo nano ultralcd.py**
9. To exit from the program press **Ctrl-Z.**
10. When we run the program, ultrasonic sensor will show the distance of obstacle came in front of it and this distance displays on terminal window as well as LCD.

EXPERIMENT NO. 8: RGB WITH RASPBERRY PI

AIM: To study interfacing of RGB with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 6 pin connector, 5v, 3A power supply.

INTRODUCTION: - RGB LED means red, blue and green LEDs. RGB LED products combine these three colors to produce over 16 million hues of light. Some colors are "outside" the triangle formed by the RGB LEDs.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 5 pin connector from the board to "RGB" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes.**
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**To run this specific program enter the command >> **rgb.py**
8. If you want to see program enter the command >> **sudo nano rgb.py**
9. To exit from the program press **Ctrl-Z.**
10. When we run the program, LED will emit red, green and blue colors.

EXPERIMENT NO. 9: DHT22 WITH RASPBERRY PI

AIM: To study interfacing of DHT22 with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin connector, 5v, 3A power supply.

INTRODUCTION:- The DHT22 is temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air.

PROCEDURE:

1. Connect Raspberry pi on board.Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to “DHT22” section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes.**
7. To see the list of files into Raspberry Pi sample codes folder enter command >> lsTo run this specific program enter the command >> DHT22.py
8. If you want to see program enter the command >> sudo nano DHT.py.
9. To exit from the program press Ctrl-Z.
10. Output able to see on serial terminal.

EXPERIMENT NO 10: PIR SENSOR WITH RASPBERRY PI

AIM: To study interfacing of PIR sensor with Raspberry pi.

REQUIREMENTS:- Raspberry pi ,40 pin FRC cable, 4 pin connector, 5v, 3A power supply.

INTRODUCTION: - A passive infrared sensor (PIR sensor) is an electronic sensor that measures infrared (IR) light radiating from objects in its field of view. They are most often used in PIR-based motion detectors. A PIR-based motion detector is used to sense movement of objects.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "PIR" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. To see the list of files into Raspberry Pi sample codes folder enter command >> lsTo run this specific program enter the command >> PIR.py
7. If you want to see program enter the command >> sudo nano PIR.py.
8. To exit from the program press Ctrl-Z.
9. Output able to see on serial terminal.

EXPERIMENT NO. 11: JOYSTICK WITH RASPBERRY PI

AIM: To study interfacing of Joystick with Raspberry pi.

REQUIREMENTS:- Raspberry pi ,40 pin FRC cable, 5 pin connector, 5v, 3A power supply.

INTRODUCTION:- A joystick is an input device consisting of a stick that pivots on a base and reports its angle or direction to the device it is controlling.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 5 pin connector from the board to “JOYSTICK” section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes** To see the list of files into SENSOR BOARD folder enter command >> **ls**
7. To run this specific program enter the command >> **joystick.py**
8. If you want to see program enter the command >> **sudo nano joystick.py**
9. To exit from the program press **Ctrl-Z.**
10. When you run the program, it will show the output voltages as direction in which you move the joystick. You can see the output on terminal window.

EXPERIMENT NO. 12: BLUETOOTH SENSOR WITH RASPBERRY PI

AIM: To study interfacing of Bluetooth Sensor with Raspberry pi.

REQUIREMENTS:- Raspberry pi ,40 pin FRC cable,4 pin connector, 5v,3A power supply. **INTRODUCTION:-** A Bluetooth device works by using radio waves instead of wires or cables to connect with your Smartphone or computer. Bluetooth is a wireless short-range communications technology standard found in millions of products.

PROCEDURE:

1. Connect Raspberry pi on board. Connect TYPE C power cable to PI.
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "BLUETOOTH" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes**
7. To see the list of files into **Raspberry Pi sample codes** folder enter command >> **ls**To run this specific program enter the command >> **bluetooth.py**
8. If you want to see program enter the command >> **sudo nano bluetooth.py**
9. To exit from the program press **Ctrl-Z.**
10. Download any Bluetooth Application (E.g. Blue Chat) in your mobile device. Bluetooth sensor on board will connect to your mobile phone's Bluetooth and when you send any message from mobile using that application, you will get that message on your terminal window.

EXPERIMENT NO. 13: WI-FI WITH RASPBERRY PI

AIM: To study interfacing of WI-FI with Raspberry pi.

REQUIREMENTS: - Raspberry pi, 40 pin FRC cable, 4 pin connector, 5v, 3A power supply.

INTRODUCTION: - The wireless sensor tag is a real-time, cloud based system, so any change you make from one device instantly appears on your other devices.

PROCEDURE:

1. Connect Raspberry pi on board Connect TYPE C power cable to PI. .
2. Connect the HDMI screen, mouse and keyboard to Raspberry pi.
3. Switch ON the power supply.
4. Connect 40 pin FRC cable from Raspberry pi to the 40 pin connector which is placed on the board. And connect 4 pin connector from the board to "WI-FI" section.
5. Open the Terminal window of Raspberry pi and enter the command >>
   >> **cd /home/pi/Raspberry Pi sample codes.**
6. Now we are into the folder **Raspberry Pi sample codes**
7. To see the list of files into **Raspberry Pi sample codes** folder enter command
   >> **ls**
8. To run this specific program enter the command >> **wifi.py**
9. If you want to see program enter the command >> **sudo nano wifi.py**
10. To exit from the program press **Ctrl-Z.**
11. Go to terminal window and write AT on it, you will get the reply as AT OK for startup, AT+RST for restart and AT+GMR to view version information.

    This means Wi-Fi is connected successfully.

CHAPTER 5. FRC CONNECTOR DETAILS FOR RASPBERRY PI

1. 40 PIN FRC CONNECTOR (J9):

Pin No.	Description	Pin No.	Description
1	NC	2	VDD_5V
3	SDA	4	VDD_5V
5	SCL	6	GND
7	NC	8	RXGPIO_14
9	GND	10	TXGPIO_15
11	NC	12	GPIO_18
13	GPIO_27	14	GND
15	GPIO_22	16	GPIO_23
17	NC	18	GPIO_24
19	GPIO_10	20	GND
21	GPIO_9	22	GPIO_25
23	GPIO_11	24	GPIO_8
25	GND	26	GPIO_7
27	NC	28	NC
29	GPIO_5	30	GND
31	GPIO_6	32	GPIO_12
33	GPIO_13	34	GND
35	GPIO_19	36	GPIO_16
37	GPIO_26	38	GPIO_20
39	GND	40	GPIO_21

2. 8 PIN CONNECTOR FOR KEYPAD (RJ1)

Pin No.	Description
1	GIO_4(GPIO_GCLK)
2	GIO_5
3	GIO_6
4	GIO_7(SPI_CE1_N)
5	GIO_8(SPI_CE0_N)
6	GIO_9(SPI_MISO)
7	GIO_10(SPI_MOSI)
8	GIO_11(SPI_CLK)

3. 8 PIN CONNECTOR FOR LCD (RJ2)

Pin No.	Description
1	5/3V
2	GIO_13
3	GIO_14(TXD0)
4	GIO_15(RXD0)
5	GIO_16
6	GIO_17(GPIO_GEN0)
7	GIO_18(GPIO_GEN1)
8	GND

4. 8 PIN CONNECTOR FOR STEPPER (RJ3)

Pin No.	Description
1	5/3V
2	GIO_19
3	GIO_20
4	GIO_21
5	GIO_22(GPIO_GEN3)
6	GIO_23(GPIO_GEN4)
7	GIO_24(GPIO_GEN5)
8	GND

5. 6 PIN CONNECTOR FOR RGB (RJ4)

Pin No.	Description
1	GIO_19
2	GIO_21
3	GIO_22(GPIO_GEN3)
4	GIO_23(GPIO_GEN4)
5	GIO_24(GPIO_GEN5)
6	GND

6. 4 PIN CONNECTOR FOR ULTRASONIC (RJ5)

Pin No.	Description
1	5/3V
2	GIO_21
3	GIO_22(GPIO_GEN3)
4	GND

7. 4 PIN CONNECTOR FOR BLUETOOTH (RJ11)

Pin No.	Description
1	5/3V
2	GPIO_14(TXD0)
3	GPIO_15(RXT0)
4	GND

8. 4 PIN CONNECTOR FOR SERVO MOTOR (RJ5)

Pin No.	Description
1	5/3V
2	GPIO_21
3	GPIO_22
4	GND

9. 4 PIN CONNECTOR FOR PIR SENSOR(RJ5)

Pin No.	Description
1	5/3V
2	GPIO_21
3	GPIO_22
4	GND

10. 4 PIN CONNECTOR FOR DHT22 (RJ6)

Pin No.	Description
1	5/3V
2	GPIO_4
3	GPIO_5
4	GND

11. 4 PIN CONNECTOR FOR RELAY-BUZZER (RJ5)

Pin No.	Description
1	5/3V
2	GPIO_21
3	GPIO_22
4	GND

12. 4 PIN CONNECTOR FOR LCD-I2C (RJ12)

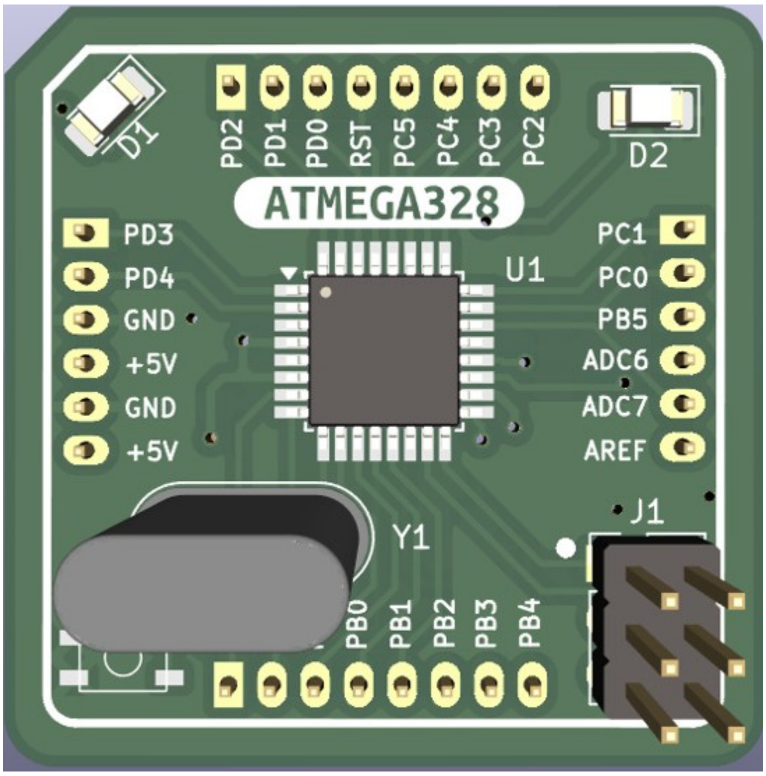
Pin No.	Description
1	5/3V
2	GPIO_2
3	GPIO_3
4	GND

CHAPTER 6. INTRODUCTION TO ARDUINO UNO (ATMEGA328P)

6.1 INTRODUCTION TO ARDUINO UNO (ATMEGA328P)

The Arduino Uno is a foundational microcontroller board in the world of embedded systems and the Internet of Things (IoT). Powered by the ATmega328P microcontroller, it provides a simple and cost-effective platform for developing digital systems that interact with the physical world. The board is widely adopted by hobbyists, students, researchers, and engineers for its ease of use, robust hardware, and vast open-source ecosystem.

In the RASPDUINO platform—where both Raspberry Pi and Arduino Uno are integrated—the Arduino acts as the hardware-level interface for sensors and actuators. It can perform real-time tasks, collect sensor data, and control devices, while the Raspberry Pi handles more advanced processing, networking, or cloud integration. This hybrid model allows learners and developers to create scalable and modular IoT applications.



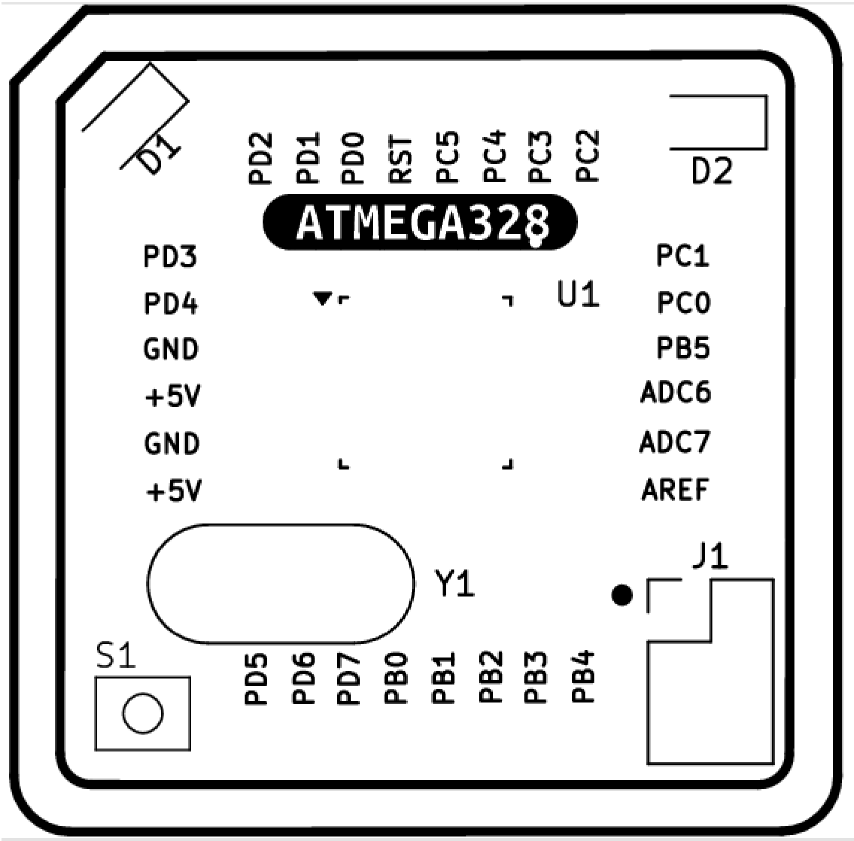
Arduino Uno

The **Arduino Uno** is a microcontroller board based on the ATmega328. Arduino is an open-source, prototyping platform and its simplicity makes it ideal for hobbyists to use as well as professionals. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC- to-DC adapter or battery to get started.

6.2 KEY FEATURES OF ARDUINO UNO (ATMEGA328P)

- **Microcontroller: ATmega328P, 8-bit AVR**
- **Operating Voltage: 5V**
- **Input Voltage (recommended): 7–12V**
- **Digital I/O Pins: 14 (of which 6 provide PWM output)**
- **Analog Input Pins: 6**
- **Flash Memory: 32 KB (of which 0.5 KB is used by the bootloader)**
- **SRAM: 2 KB**
- **EEPROM: 1 KB**
- **Clock Speed: 16 MHz**
- **USB Interface: For programming and serial communication**
- **Open-source IDE: Arduino IDE with easy-to-learn C/C++-based syntax**

6.3 ARDUINO UNO (ATMEGA328P) BOARD COMPONENT LEGEND DIAGRAM


D1
PD2
PD1
PD0
RST
PC5
PC4
PC3
PC2
D2
ATMEGA328
PD3
PD4
GND
+5V
GND
+5V
U1
PC1
PC0
PB5
ADC6
ADC7
AREF
Y1
J1
S1
PD5
PD6
PD7
PB0
PB1
PB2
PB3
PB4

Specification Details:

Specification	Details
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7–12V
Input Voltage (limits)	6–20V
Digital I/O Pins	14 (of which 6 provide PWM output)
PWM Digital I/O Pins	6 (Pins 3, 5, 6, 9, 10, 11)
Analog Input Pins	6 (A0 to A5)
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (0.5 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Clock Speed	16 MHz
LED_BUILTIN	Pin 13
USB Connection	USB Type-B (for power and programming)
Power Jack	Barrel jack (7–12V input recommended)
ICSP Header	Yes
Reset Button	Yes
Dimensions	68.6 mm × 53.4 mm
Weight	Approx. 25 g

6.4 PINOUT OVERVIEW

The Arduino Uno board offers a user-friendly layout of pins that support various input and output functions:

- **Digital Pins (0–13):** Used for digital input/output, including PWM outputs on pins 3, 5, 6, 9, 10, and 11.
- **Analog Pins (A0–A5):** Used for reading analog signals from sensors.
- **Power Pins:** Includes 5V, 3.3V, GND, and Vin.
- **Communication Pins:**
  - UART (TX/RX): Pins 0 and 1
  - SPI: Pins 10 (SS), 11 (MOSI), 12 (MISO), and 13 (SCK)
  - I2C: A4 (SDA) and A5 (SCL)
- **Reset Pin:** Allows external reset of the microcontroller

CHAPTER 7. SETTING FOR ARDUINO UNO TO USE AS DEVELOPMENT BOARD

INSTALL Arduino 1.8.3 –

The open source arduino software (IDE) makes it easy to write code and upload it to the board. It runs on Windows, MAC os x, Linux. This software can be used with any arduino board.

DOWNLOAD-

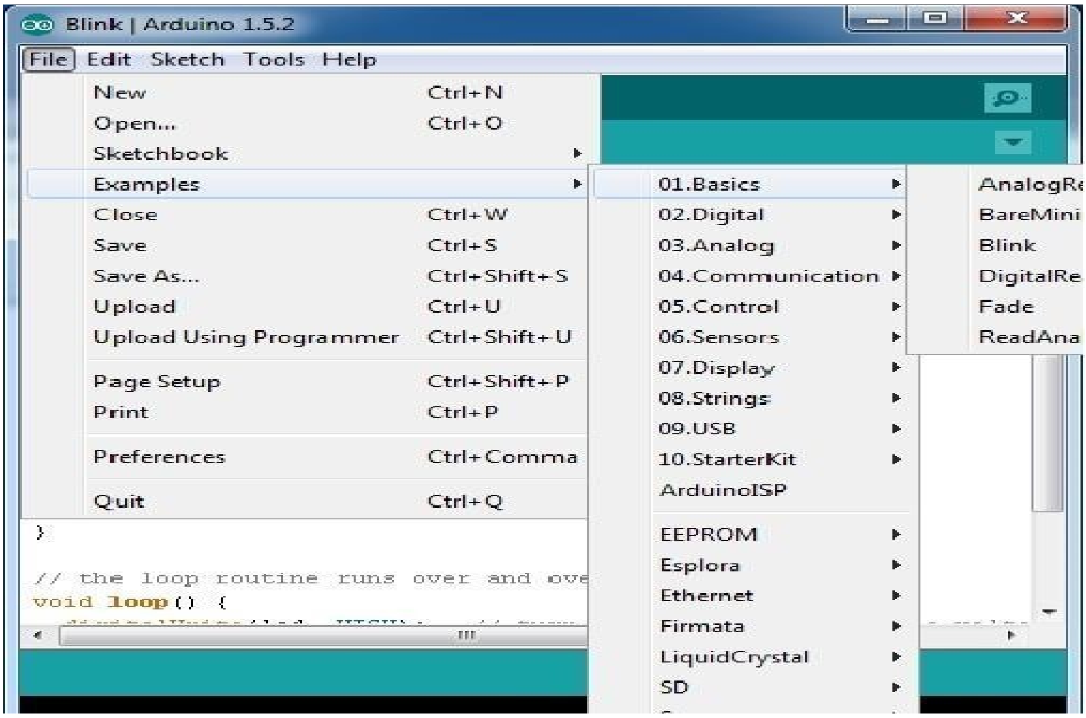
Download the software from-

https://www.arduino.cc/en/Mai
n/Software

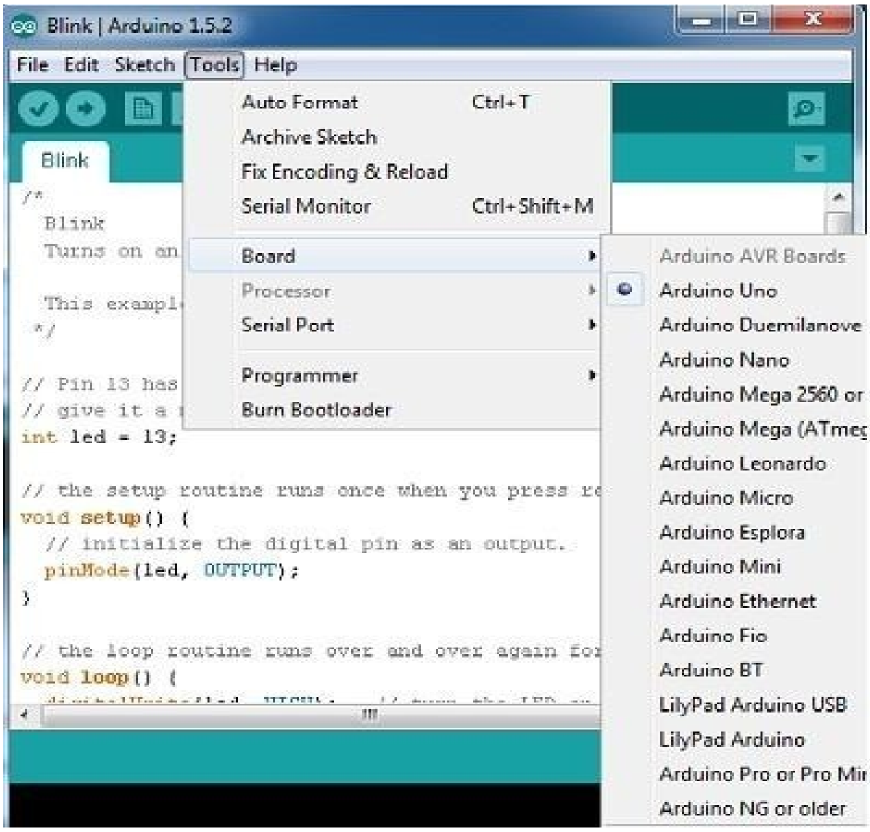
Launch and Blink-

After following the appropriate steps for your software install, we are now ready to test your first program with your Arduino board!

- Launch the Arduino application
- If you disconnected your board, plug it back in
- Open the Blink example sketch by going to: File > Examples > 1.Basics > Blink

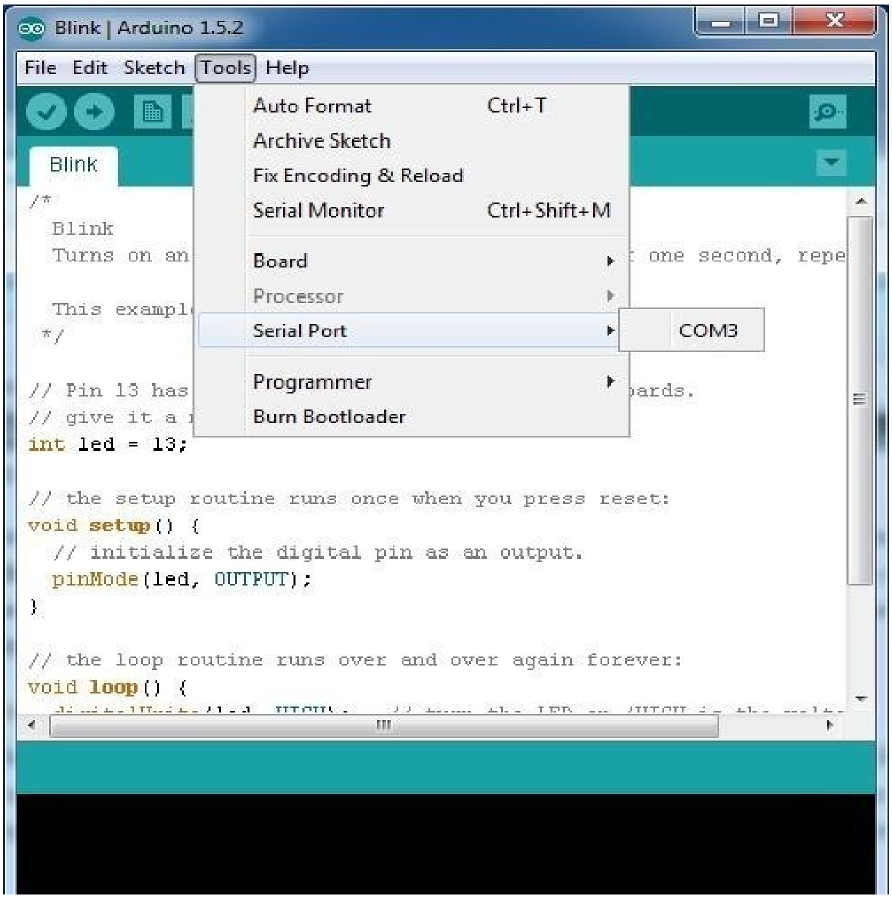


- Select the type of Arduino board you're using: Tools > Board > your board type

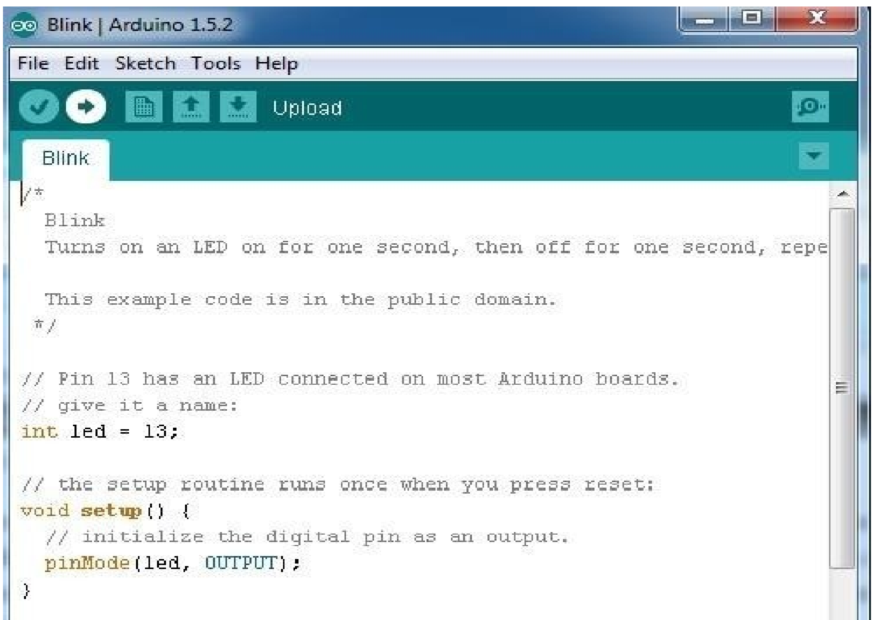


- Select the serial/COM port that your Arduino is attached to: Tools > Port >

COMxx



- If you’re not sure which serial device is your Arduino, take a look at the available ports, then unplug your Arduino and look again. The one that disappeared is your Arduino.
- With your Arduino board connected, and the Blink sketch open, press the ‘Upload’ button



- After a second, you should see some LEDs flashing on your Arduino, followed by the message ‘Done Uploading’ in the status bar of the Blink sketch.
- If everything worked, the on-board LED on your Arduino should now be blinking! You just programmed your first Arduino!

CHAPTER 8. EXPERIMENTS OF ARDUINO UNO

EXPERIMENT NO. 1: LCD DISPLAY

AIM: To study the interfacing of LCD (16*2) with the **Arduino ATmega328.**

REQUIREMENTS: Arduino, 8 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file: open particular ".ino" file of LCD program.
4. Compile the program and then upload it.
5. After uploading connect 8 pin connector from the board to "LCD" section.
6. You will observe that it displays "LOGSUN SYSTEMS" on LCD display.

EXPERIMENT NO. 2: I2C LCD DISPLAY

AIM: To study the interfacing of I2C LCD (16*2) with the **Arduino ATmega328.**

REQUIREMENTS: Arduino, 4 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of lcd_i2c program.
4. Compile the program and then upload it.
5. After uploading, connect 4 pin connector from the board to "LCD" section.
6. You will observe that it displays "LOGSUN SYSTEMS, I2C DEMO" on LCD display.

EXPERIMENT NO. 3: LCD WITH KEYPAD INTERFACING

AIM: To study the interfacing of LCD and Keypad with **Arduino ATmega328..**

REQUIREMENTS: Arduino, two 8 pin connectors, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file: open particular ".ino" file of LCD Keypad program.
4. Compile the program and then upload it.
5. After uploading connect 8 pin connector from the board to "LCD-KEYPAD" section.
6. You will observe that when you press a key (0-F) it will show on LCD display.

EXPERIMENT NO. 4: INTERFACING STEPPER MOTOR WITH ARDUINO

AIM: To study the interfacing of stepper motor to **Arduino ATmega328..**

REQUIREMENTS: Arduino, 8 pin connector, 12V power supply or USB cable, Stepper motor, 12v dc external power supply.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of Stepper motor program.
4. Compile the program and then upload it.
5. After uploading connect 8 pin connector from the board to "STEPPER MOTOR" section.
6. You will observe that the motor will starts automatically and after some time it slows down and again its speed increases.

EXPERIMENT NO. 5: INTERFACING SERVO MOTOR WITH ARDUINO

AIM: To study the interfacing of stepper motor to **Arduino ATmega328.**

REQUIREMENTS: Arduino, 4 pin connector, 12V power supply or USB cable, Servo motor.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of Servo motor program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "SERVO-MOTOR" section.
6. You will observe that the motor will rotate in 0-90-180-360 degrees

---

EXPERIMENT NO. 6: INTERFACING RELAY-BUZZER WITH ARDUINO

AIM: To study interfacing of relay with **Arduino ATmega328.**

REQUIREMENTS: - Arduino, 4 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from AJ10 to the board to "RELAY-BUZZER" section.
6. You will observe that relay is ON for 5sec and OFF for 5 sec and buzzer will also ON at a time

EXPERIMENT NO. 7: INTERFACING ULTRASONIC SENSOR WITH ARDUINO

AIM: To study interfacing ultrasonic sensor with **Arduino ATmega328.**

REQUIREMENTS: - Arduino, 4 and 8 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "ULTRASONIC" section and 8 pin connector to "LCD" section.
6. When we run the program, ultrasonic sensor will show the distance of obstacle came in front of it and this distance displays on LCD.

EXPERIMENT NO. 8: INTERFACING RGB WITH ARDUINO

AIM: To study interfacing of RGB with **Arduino ATmega328.**

REQUIREMENTS:- Arduino, 5 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "RGB" section.
6. You will observe that LED will emit red, green and blue colors.

EXPERIMENT NO. 9: INTERFACING DHT22 WITH ARDUINO

AIM: To study interfacing of DHT22 with **Arduino ATmega328.**

REQUIREMENTS:- Arduino, 4 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "DHT22" section.
6. You can observe that DHT22 sensor will show the temperature and humidity on LCD display.

EXPERIMENT NO. 10: INTERFACING PIR SENSOR WITH ARDUINO

AIM: To study interfacing of PIR with **Arduino ATmega328.**

REQUIREMENTS:- Arduino, 4 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file: open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "PIR" section.

You can observe that PIR sensor detects motion in its field and it shows on terminal window whether the motion is detected or not

---

EXPERIMENT NO. 11: INTERFACING JOYSTICK WITH ARDUINO

AIM: To study interfacing of Joystick with **Arduino ATmega328.**

REQUIREMENTS: - Arduino, 5 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file: open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 5 pin connector from the board to "JOYSTICK" section.
6. It will show the output voltages as direction in which you move the joystick. You can see the output on terminal window

---

EXPERIMENT NO. 12: INTERFACING OF BLUETOOTH WITH ARDUINO

AIM: To study interfacing of Bluetooth with **Arduino ATmega328.**

REQUIREMENTS: - Arduino, 4 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "BLUETOOTH" section.
6. Download any Bluetooth Application (E.g. Blue Chat) in your mobile device.

Bluetooth sensor on board will connect to your mobile phone's Bluetooth and when you send any message from mobile using that application, you will get that message on your terminal window.

EXPERIMENT NO. 13: INTERFACING OF WI-FI WITH ARDUINO

AIM: To study interfacing of WI-FI with **Arduino ATmega328.**

REQUIREMENTS:- Arduino, 4 pin connector, 12V power supply or USB cable.

PROCEDURE:

1. Open the Arduino IDE. Connect USB cable to Arduino Uno.
2. Select Arduino Uno and COM port from TOOL section.
3. Now go to file : open particular ".ino" file of relay program.
4. Compile the program and then upload it.
5. After uploading connect 4 pin connector from the board to "WI-FI" section.
6. Go to terminal window and write AT on it, you will get the reply as AT OK for startup, AT+RST for restart module, AT+GMR for version information. This means Wi-Fi is connected successfully.

CHAPTER 9. PIN CONNECTOR DETAILS

1. 8 PIN CONNECTOR FOR KEYPAD (AJ1)

Pin No.	Description
1	IO_0
2	IO_1
3	IO_2
4	IO_3
5	IO_4
6	IO_5
7	IO_6
8	IO_7

2. 8 PIN CONNECTOR FOR LCD (AJ2)

Pin No.	Description
1	5/3V
2	IO_13
3	IO_12
4	IO_11
5	IO_10
6	IO_9
7	IO_8
8	GND

3. 8 PIN CONNECTOR FOR STEPPER(AJ3)

Pin No.	Description
1	5/3V
2	IO_13
3	IO_12
4	IO_11
5	IO_10
6	IO_9
7	IO_8
8	GND

4. PIN CONNECTOR FOR RGB(AJ4)

Pin No.	Description
1	AD_4
2	AD_3
3	AD_2
4	AD_1
5	AD_0
6	GND

5. PIN CONNECTOR FOR JOYSTICK(AJ5) to J6

Pin No.	Description
1	5/3V
2	AD_0
3	AD_1
4	IO_2
5	GND

6. **4 PIN CONNECTOR FOR LCD_I2C(AJ12)**

Pin No.	Description
1	5/3V
2	AD_4
3	AD_5
4	GND

7. **4 PIN CONNECTOR FOR SERVO(AJ6)**

Pin No.	Description
1	5/3V
2	AD_3
3	AD_2
4	GND

8. **4 PIN CONNECTOR FOR RELAY_BUZZER(AJ9)**

Pin No.	Description
1	5/3V
2	IO_6
3	IO_7
4	GND

9. **4 PIN CONNECTOR FOR DHT22(AJ10)**

Pin No.	Description
1	5/3V
2	IO_6
3	IO_7

4	GND

10. 4 PIN CONNECTOR FOR BLUETOOTH(AJ10)

Pin No.	Description
1	5/3V
2	IO_6
3	IO_7
4	GND

11. 4 PIN CONNECTOR FOR WI-FI (AJ10)

Pin No.	Description
1	5/3V
2	IO_6
3	IO_7
4	GND

12. 4 PIN CONNECTOR FOR ULTRASONIC(AJ11)

Pin No.	Description
1	5/3V
2	IO_6
3	IO_7
4	GND

13. 4 PIN CONNECTOR FOR PIR(AJ10)

XIV. Pin No.	Description
1	5/3V
2	IO_6

3	IO_7
4	GND

CHAPTER 10. THINGSBOARD USER GUIDE

10.1 WHAT IS THINGSBOARD?

ThingsBoard is an open-source IoT platform that enables rapid development, management, and scaling of IoT projects. Our goal is to provide the out-of-the-box IoT cloud or on-premises solution that will enable server-side infrastructure for your IoT applications.

Features

With ThingsBoard, you are able to:

- Provision devices, assets and customers, and define relations between them.
- Collect and visualize data from devices and assets.
- Analyze incoming telemetry and trigger alarms with complex event processing.
- Control your devices using remote procedure calls (RPC).
- Build work-flows based on a device life-cycle event, REST API event, RPC request, etc.
- Design dynamic and responsive dashboards and present device or asset telemetry and insights to your customers.
- Enable use-case specific features using customizable rule chains.
- Push device data to other systems.

10.2 GETTING STARTED WITH THINGSBOARD

Upload Data to Thingsboard

Introduction

The goal of this tutorial is to demonstrate the basic usage of the most popular ThingsBoard features. You will learn how to:

- Connect devices to ThingsBoard;
- Push data from devices to ThingsBoard;

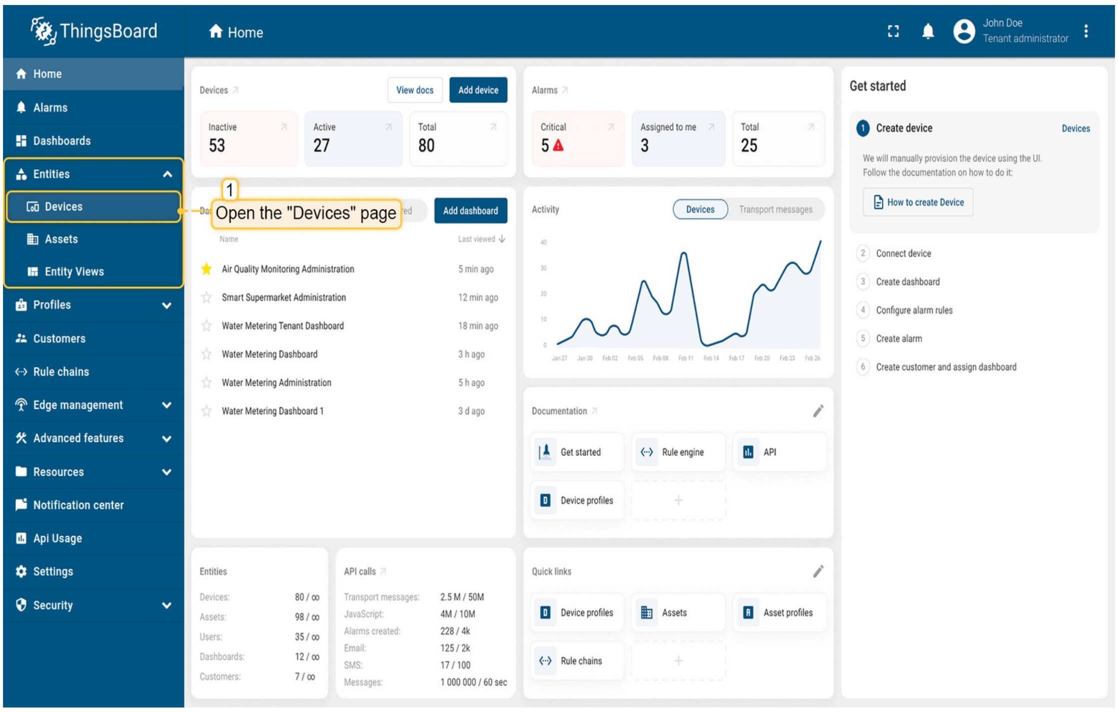
- Build real-time end-user dashboards;
- Define thresholds and trigger alarms;
- Push notifications about new alarms over email, sms or other systems.

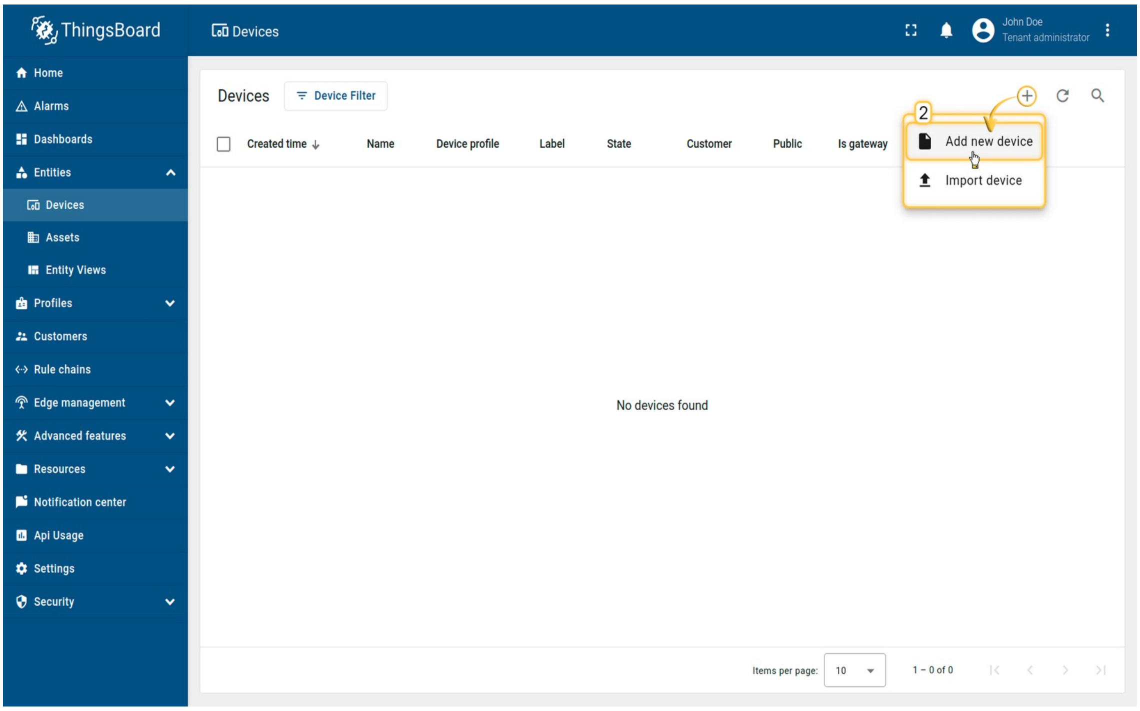
  We will connect and visualize data from the temperature sensor to keep it simple.

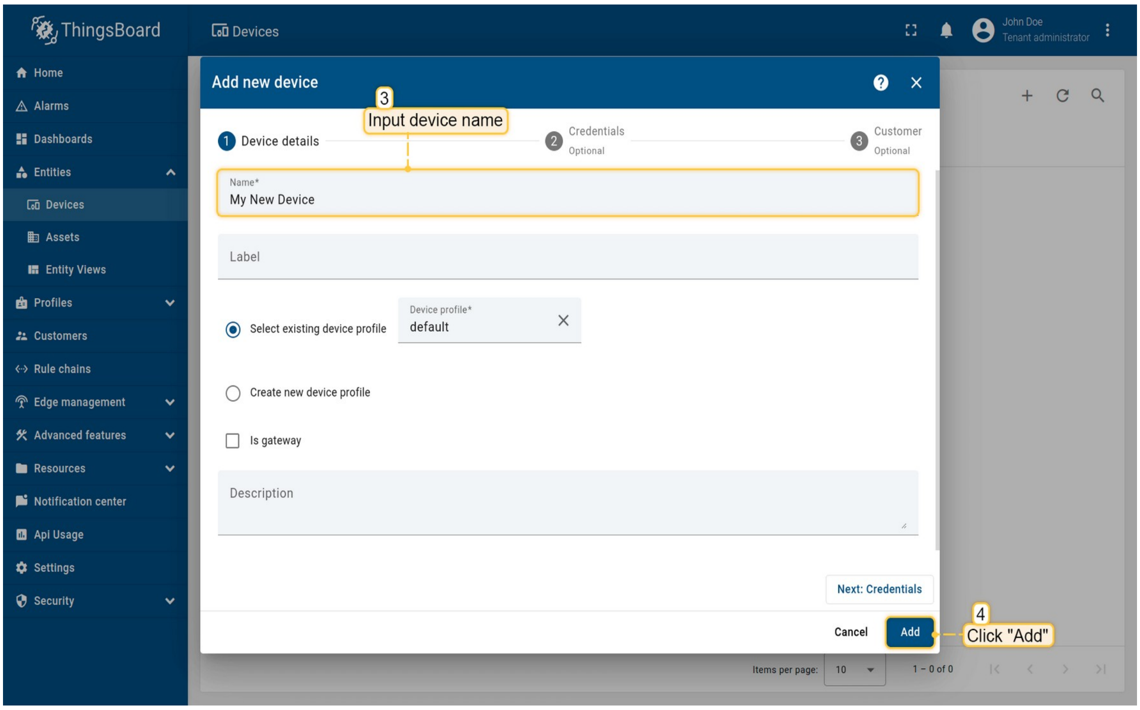
Step 1. Provision Device

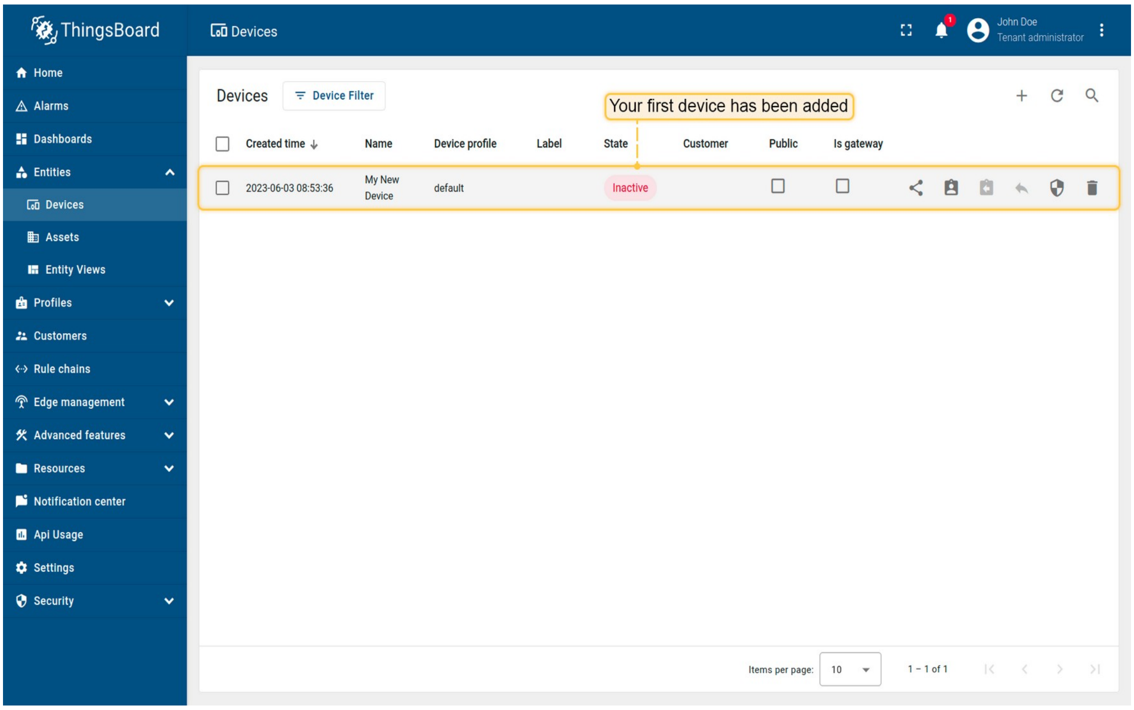
For simplicity, we will provision the device manually using the UI.

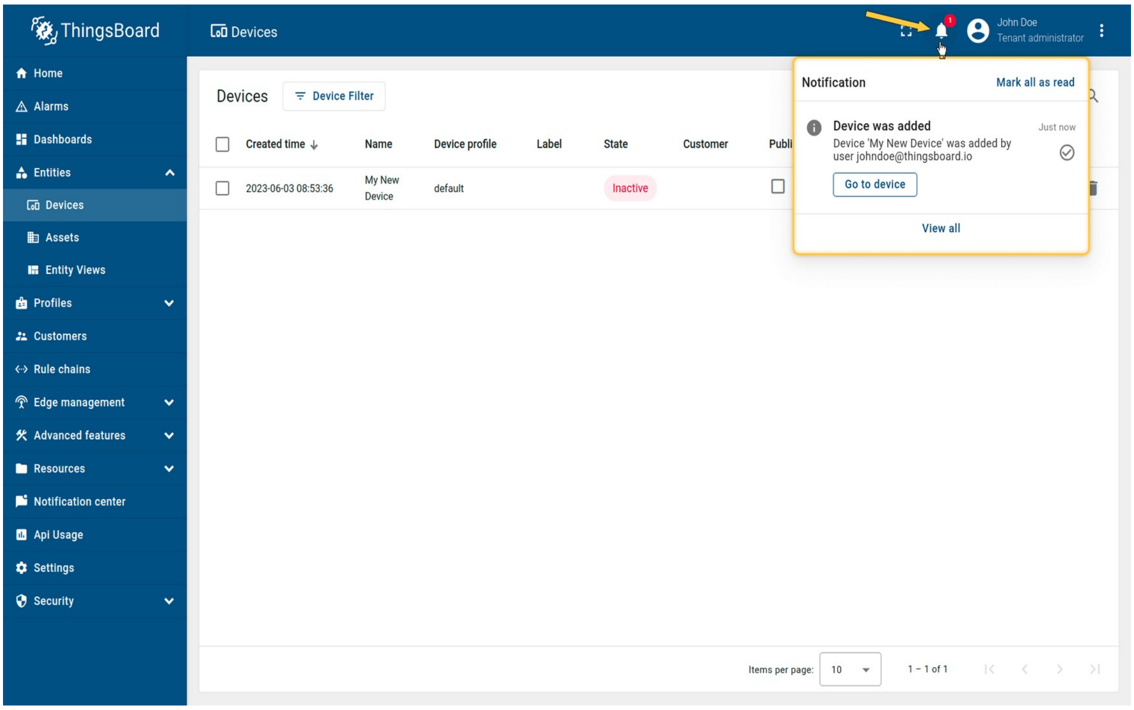
- Login to your ThingsBoard instance and navigate to the "Entities". Then click the "Devices" page;
- Click on the "+" icon in the top right corner of the table and then select "Add new device";
- Input device name. For example, "My New Device". No other changes required at this time. Click "Add" to add the device;
- Your first device has been added. As long as you have one device. But as new devices are added, they will be added to the top of the table, since the table sort devices using the time of the creation by default;
- When adding a new device, you will receive a notification. You can view it by clicking on the "bell" icon in the top right corner.




ThingsBoard
Devices
John Doe
Tenant administrator
Home
Alarms
Dashboards
Entities
Devices
Assets
Entity Views
Profiles
Customers
Rule chains
Edge management
Advanced features
Resources
Notification center
Api Usage
Settings
Security
Devices
Device Filter
Created time
Name
Device profile
Label
State
Customer
Public
Is gateway
2
Add new device
Import device
No devices found
Items per page:
10
1 – 0 of 0


ThingsBoard
Devices
John Doe
Tenant administrator
Add new device
3
Input device name
Device details
Credentials
Optional
Customer
Optional
Name*
My New Device
Label
Select existing device profile
Device profile*
default
Create new device profile
Is gateway
Description
Next: Credentials
Cancel
Add
4
Click "Add"
Items per page:
10
1 – 0 of 0

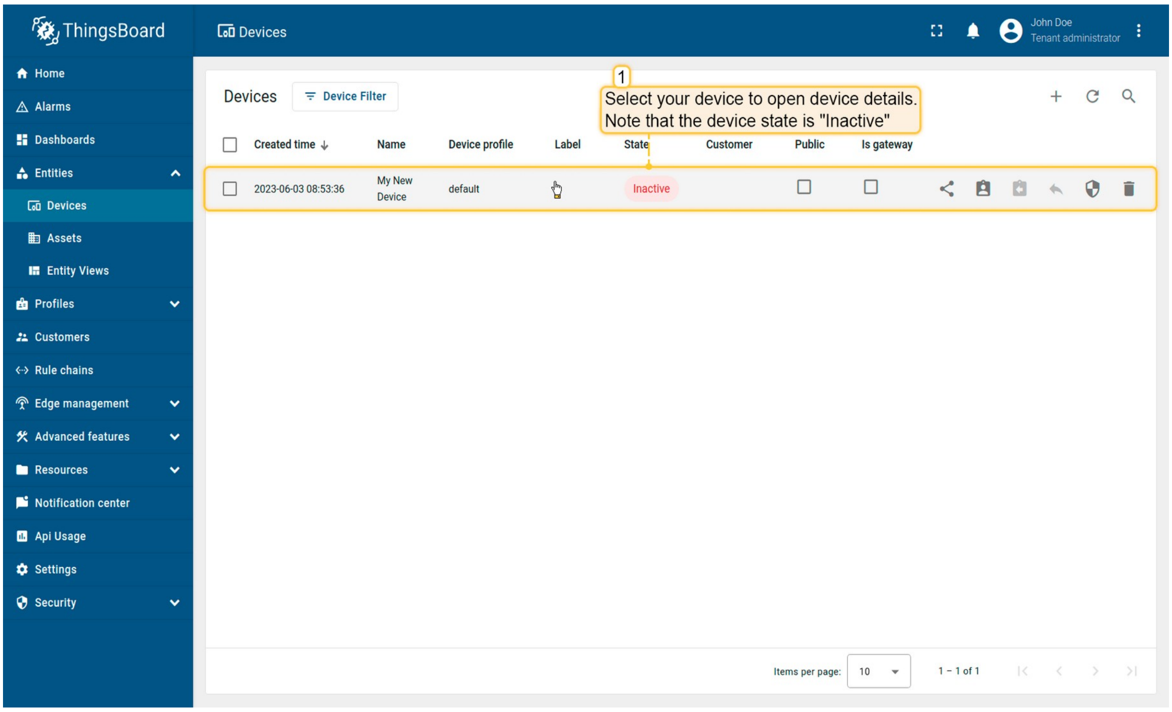


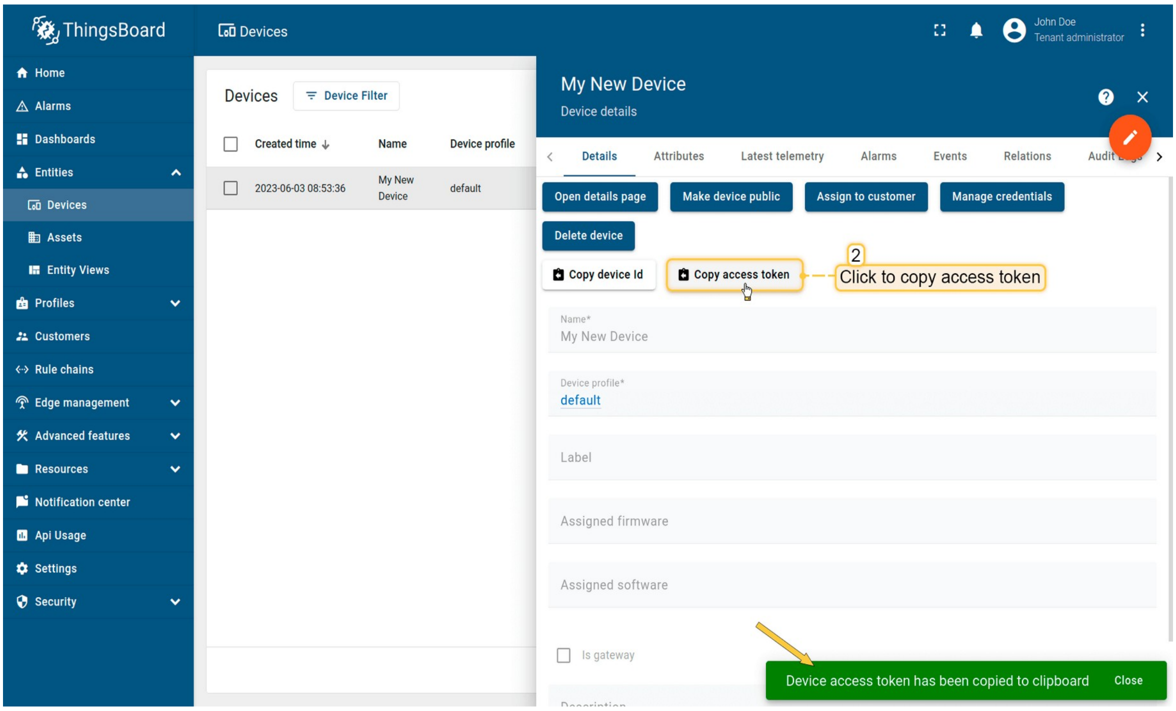


Step 2. Connect device

To connect the device you need to get the device credentials first. ThingsBoard supports various device credentials. We recommend using default auto-generated credentials which is access token for this guide.

- Click on the device row in the table to open device details. Note that the device state is "Inactive";
- Click "Copy access token". Token will be copied to your clipboard. Save it to a safe place.





Now you are ready to publish telemetry data on behalf of your device. We will use simple commands to publish data over MQTT in this DHT22 Sensor example.

Temperature upload over MQTT using Raspberry Pi and DHT22 sensor

10.3 INTRODUCTION TO THINGSBOARD

ThingsBoard is an open-source server-side platform that allows you to monitor and control IoT devices. It is free for both personal and commercial usage and you can deploy it anywhere. If this is your first experience with the platform we recommend to review what-is-thingsboard page and getting-started guide.

This sample application performs collection of temperature and humidity values produced by DHT22 sensor and further visualization on the real-time web dashboard. Collected data is pushed via MQTT to ThingsBoard server for storage and visualization. The purpose of this application is to demonstrate ThingsBoard data collection API and visualization capabilities.

The DHT22 sensor is connected to Raspberry Pi. Raspberry Pi offers a complete and self-contained Wi-Fi networking solution. Raspberry Pi push data to ThingsBoard server via MQTT protocol by using paho mqtt python library. Data is visualized using built-in customizable dashboard. The application that is running on Raspberry Pi is written in Python which is quite simple and easy to understand.

List of hardware and pinouts

- **Raspberry Pi 4**
- **DHT22 sensor**

10.4 THINGSBOARD CONFIGURATION

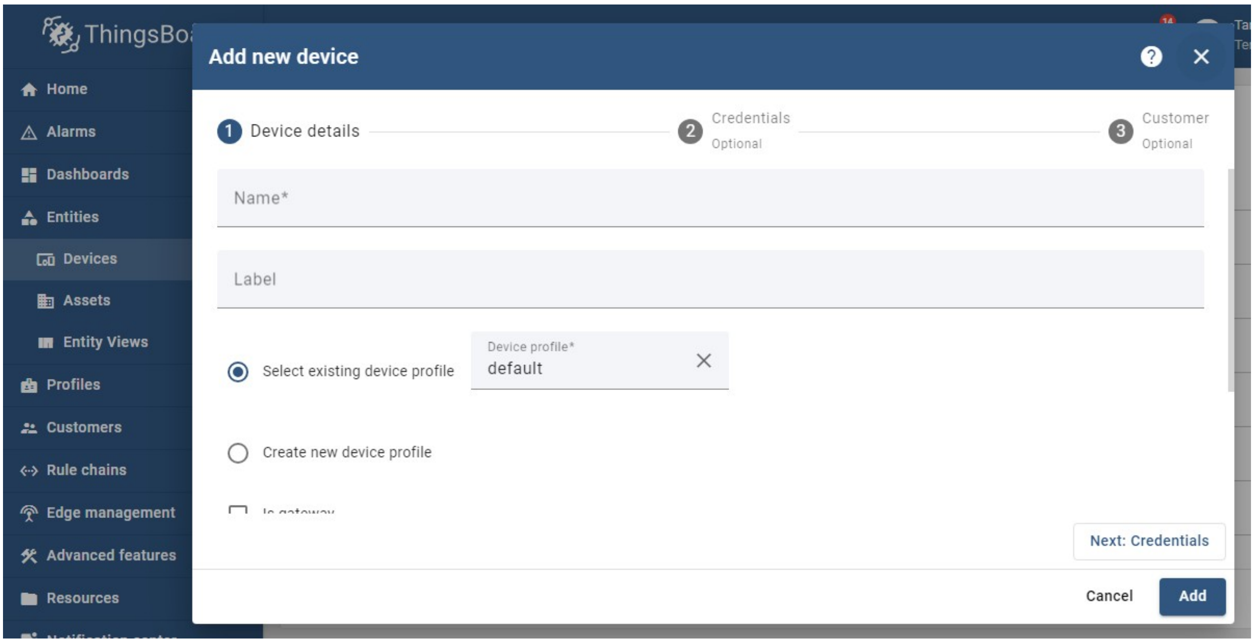
Note ThingsBoard configuration steps are necessary only in case of **local ThingsBoard installation**. If you are using Live Demo instance all entities are pre-configured for your demo account. However, we recommend reviewing this steps because you will still need to get device access token to send requests to ThingsBoard.

Provision your device

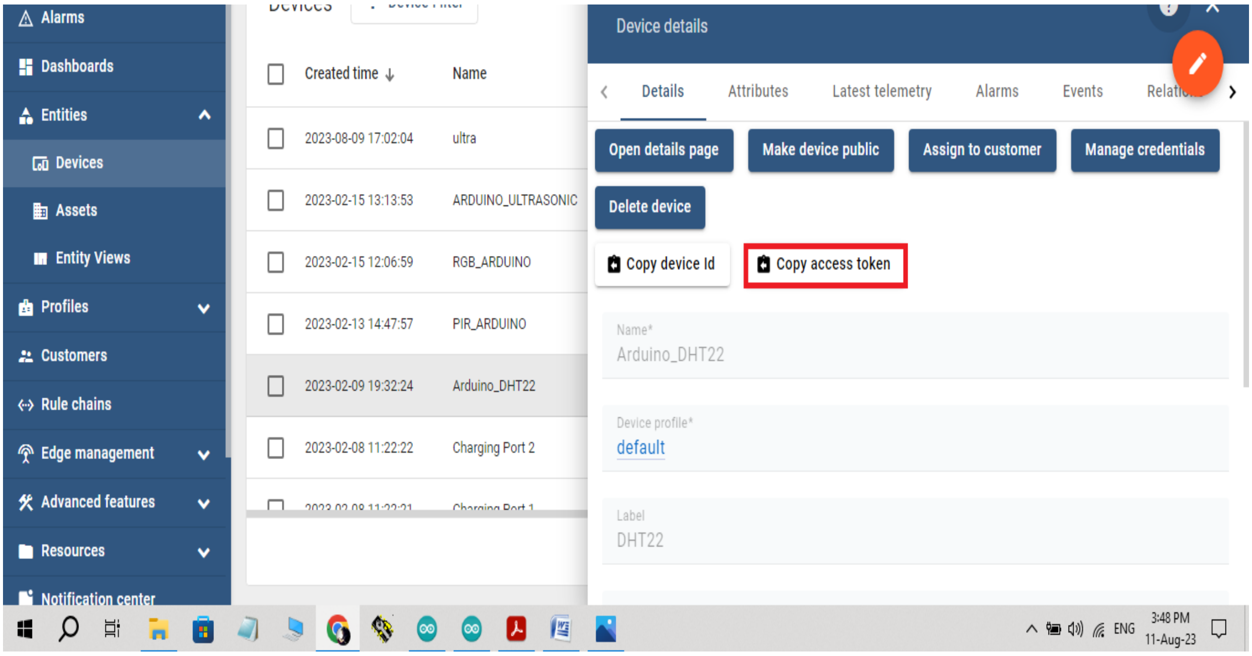
This step contains instructions that are necessary to connect your device to ThingsBoard.

Open ThingsBoarD in browser and login

Goto “Devices” section. Click “+” button and create a device with the name “DHT22 Demo Device”.



Once device created, open its details and click “Manage credentials”. Copy auto-generated access token from the “Access token” field. Please save this device token. It will be referred to later as **$ACCESS_TOKEN**.



Click "Copy access token" in device details to copy your device id to the clipboard. Paste your device id to some place, this value will be used in further steps.

10.5 PROGRAMMING THE RASPBERRY PI

Power on the raspberry pi

MQTT library installation

The following command will install MQTT Python library:

```
sudo pip install paho-mqtt
```

Adafruit DHT library installation

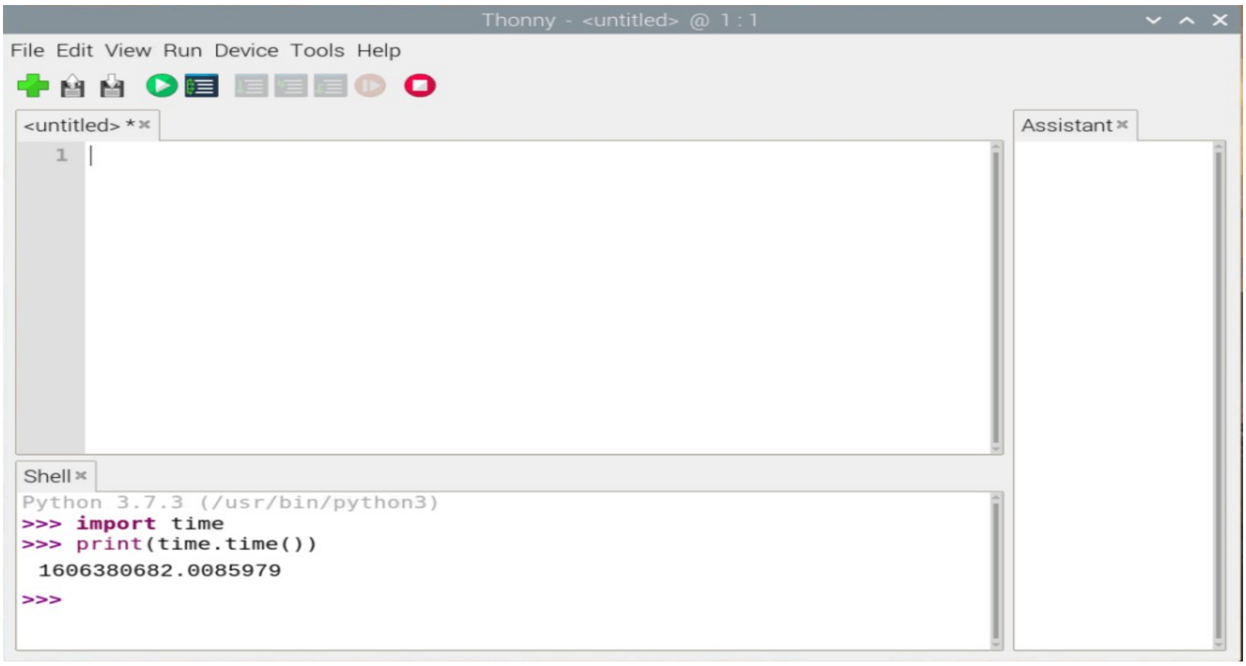
Install python-dev package:

```
sudo apt-get install python-dev
```

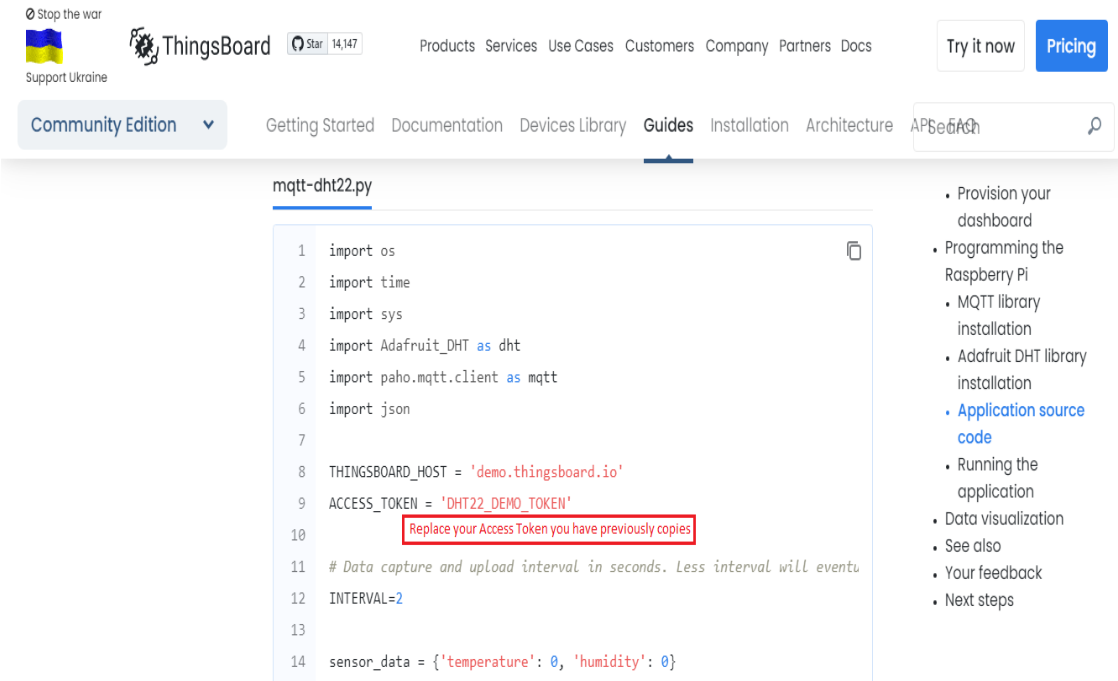
Downloading and install the Adafruit DHT library:

```
1   git clone https://github.com/adafruit/Adafruit_Python_DHT.git
2   cd Adafruit_Python_DHT
3   sudo python setup.py install
```

- Goto → home/logsun/Documents/Things-board
- Right click on dht22 code and open in Thonny IDE



Replace your access token with your copied access token previously you have done.

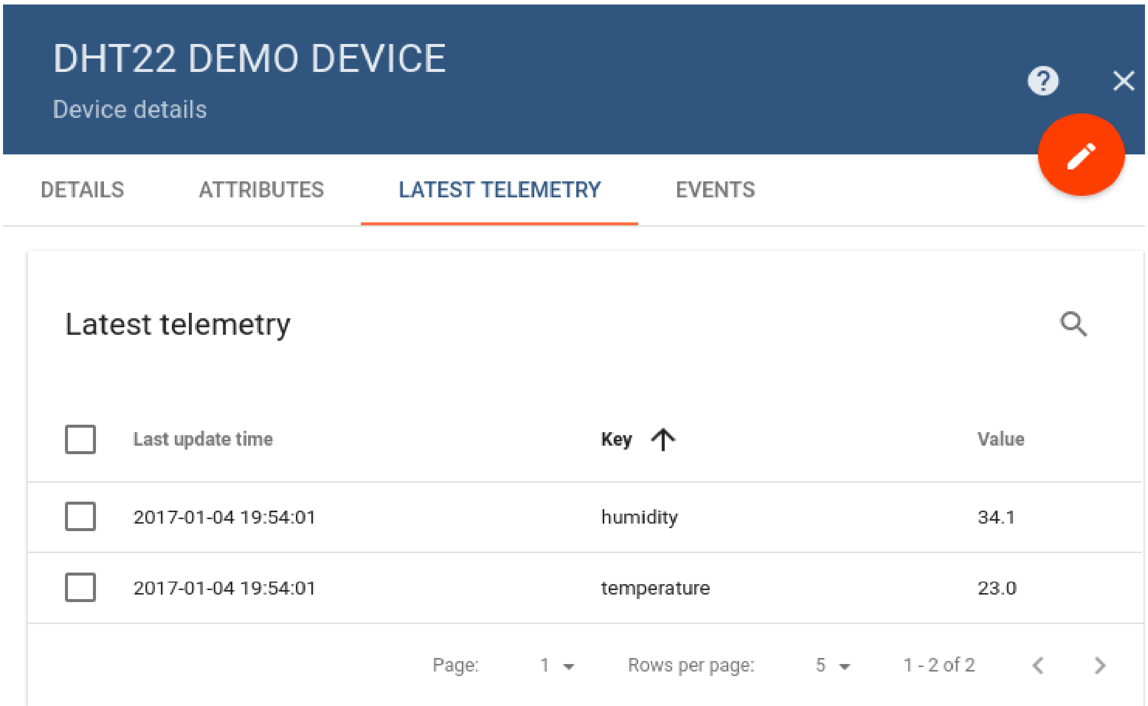


- **Click on Run to run the program**

10.6 DATA VISUALIZATION

Finally, open ThingsBoard You can access this dashboard by logging

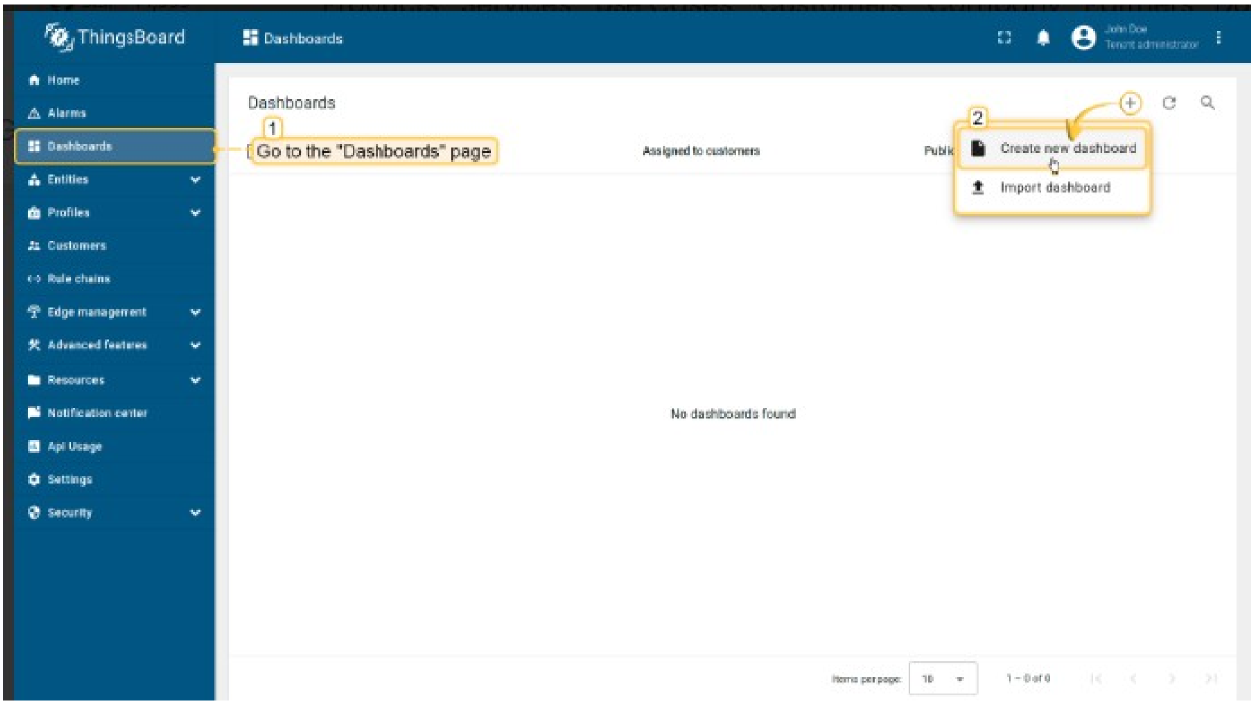
Go to **“Devices”** section and locate **“DHT22 Demo Device”**, open device details and switch to **“Latest telemetry”** tab. If all is configured correctly you should be able to see latest values of *“temperature”* and *“humidity”* in the table.

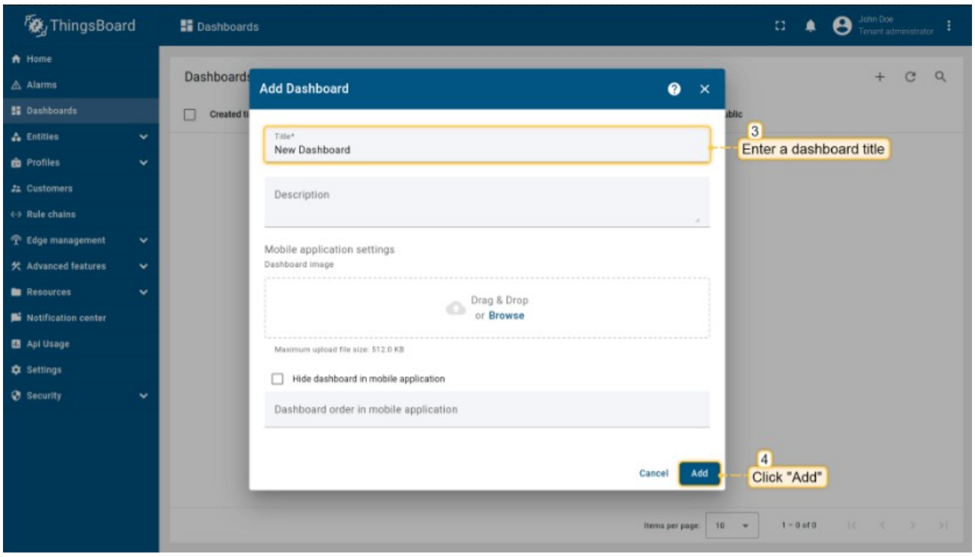


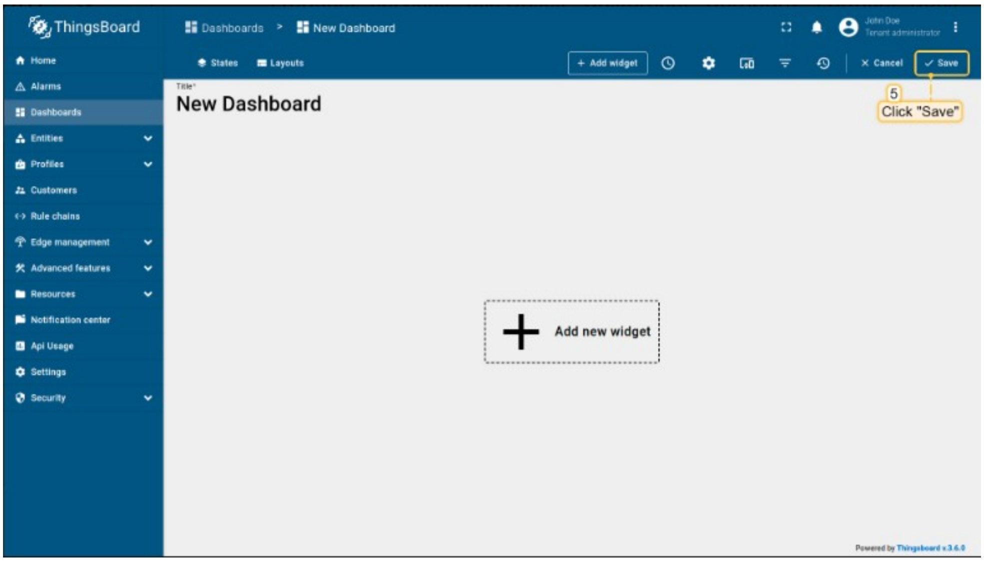
10.7 CREATE NEW DASHBOARD

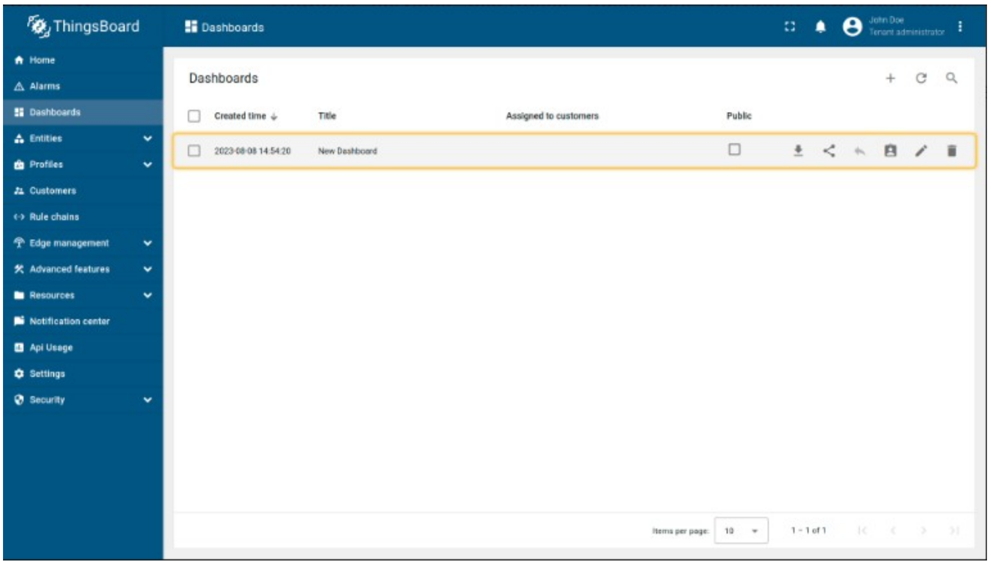
To add a new dashboard, you should:

- Login to your ThingsBoard instance and navigate to the "Dashboards" page through the main menu on the left of the screen. Then, click the "+" sign in the upper right corner of the screen, and select "Create new dashboard" from the drop-down menu;
- In the opened dialog, it is necessary to enter a dashboard title, description is optional. Click "Add";
- Once you have created the dashboard, it will be automatically opened. Save it by clicking the "Save" button in the upper right corner.
- Your first dashboard has been successfully created. As you continue to add new dashboards, they will appear at the top of the list. This default sort is based on the creation timestamp.


ThingsBoard
Dashboards
Home
Alarms
Dashboards
Entities
Profiles
Customers
Rule chains
Edge management
Advanced features
Resources
Notification center
Api Usage
Settings
Security
1
Go to the "Dashboards" page
Assigned to customers
2
Create new dashboard
Import dashboard
No dashboards found


ThingsBoard
Dashboards
Add Dashboard
New Dashboard
3
Enter a dashboard title
Description
Mobile application settings
Dashboard image
Drag & Drop
or Browse
Hide dashboard in mobile application
Dashboard order in mobile application
Cancel
Add
4
Click "Add"





10.8 EDIT DASHBOARD

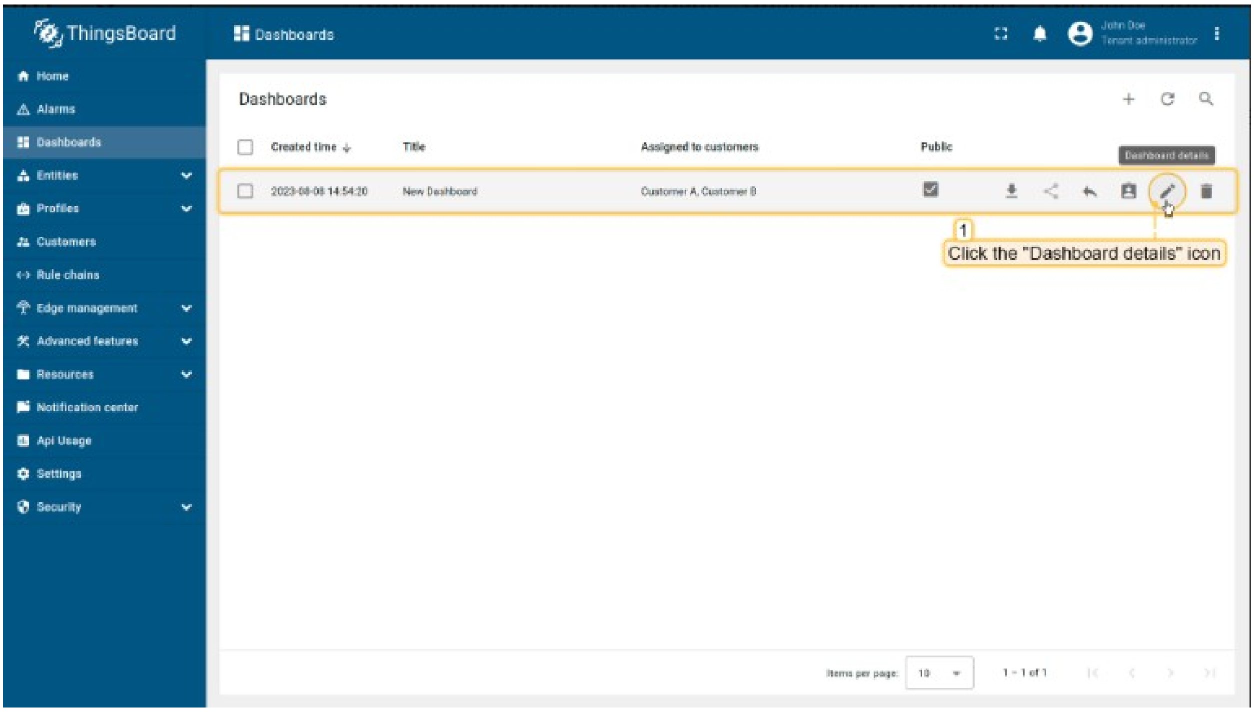
Edit the fields you need, such as the title or description of the dashboard. After making the edits, remember to save all changes. With these steps, you will have successfully updated the dashboard information.

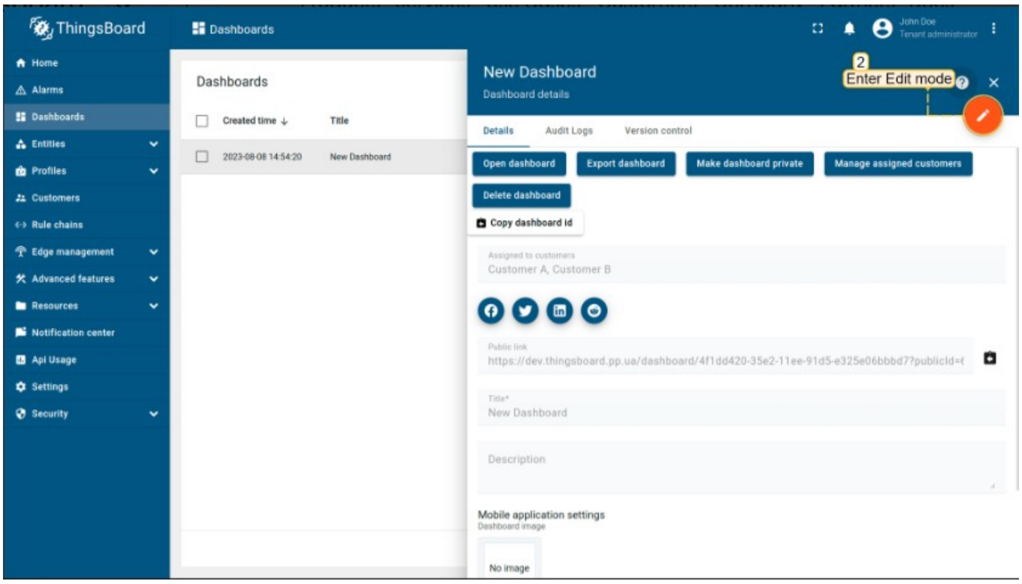
Let’s see how to do this:

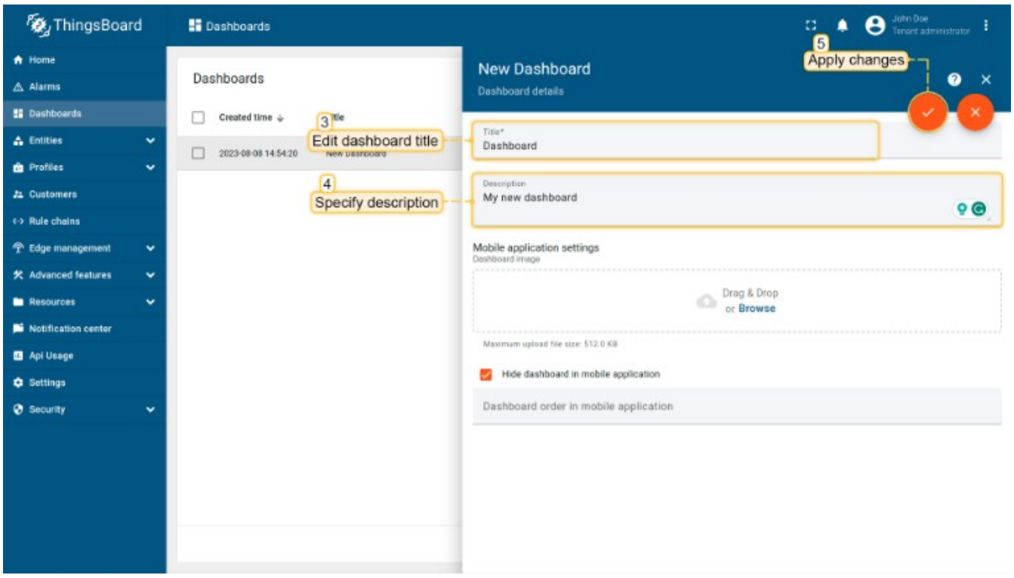
- Go to the "Dashboards" page and click the "Dashboard details" icon next to the dashboard that you want to edit;
- Click the "pencil" icon to enter edit mode;

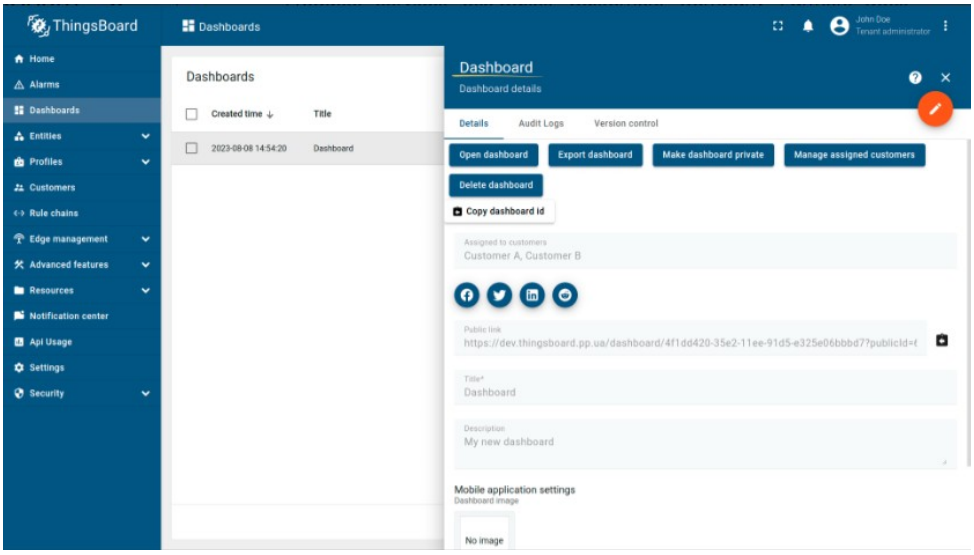
- Edit the fields you need. For example, edit the title of the dashboard, specify description. After making the edits, remember to save all changes;

- You have updated the dashboard information.









10.9 WIDGETS

10.9.1 INTRODUCTION

All IoT dashboards are constructed using ThingsBoard widgets.

A widget is an element that displays a specific type of information or functionality on a dashboard. Widgets are used to display data and visualize information obtained from devices connected to the ThingsBoard platform, remote device control, alarms management, and display static custom HTML content.

10.9.2 WIDGET TYPES

There are five types of widgets:

- **Timeseries** widgets display data for a specific time window. It can be either Realtime - the dynamically changed time frame for a certain latest interval, or History - a fixed historical time frame.
  Examples of timeseries widgets are chart widgets. Obviously, timeseries widgets are designed to display time series and not attributes;
- **Latest values** widgets display the latest values of particular attribute or time series keys. For example, device model or latest temperature reading;
- **Control** widgets allow you to send RPC commands to your devices. For example, control desired temperature on the thermostat device;
- **Alarm** widgets allow you to display alarms;
- **Static** widgets are designed to display static data. For example, floor plan or static company information.
  More about widget types you can learn here.

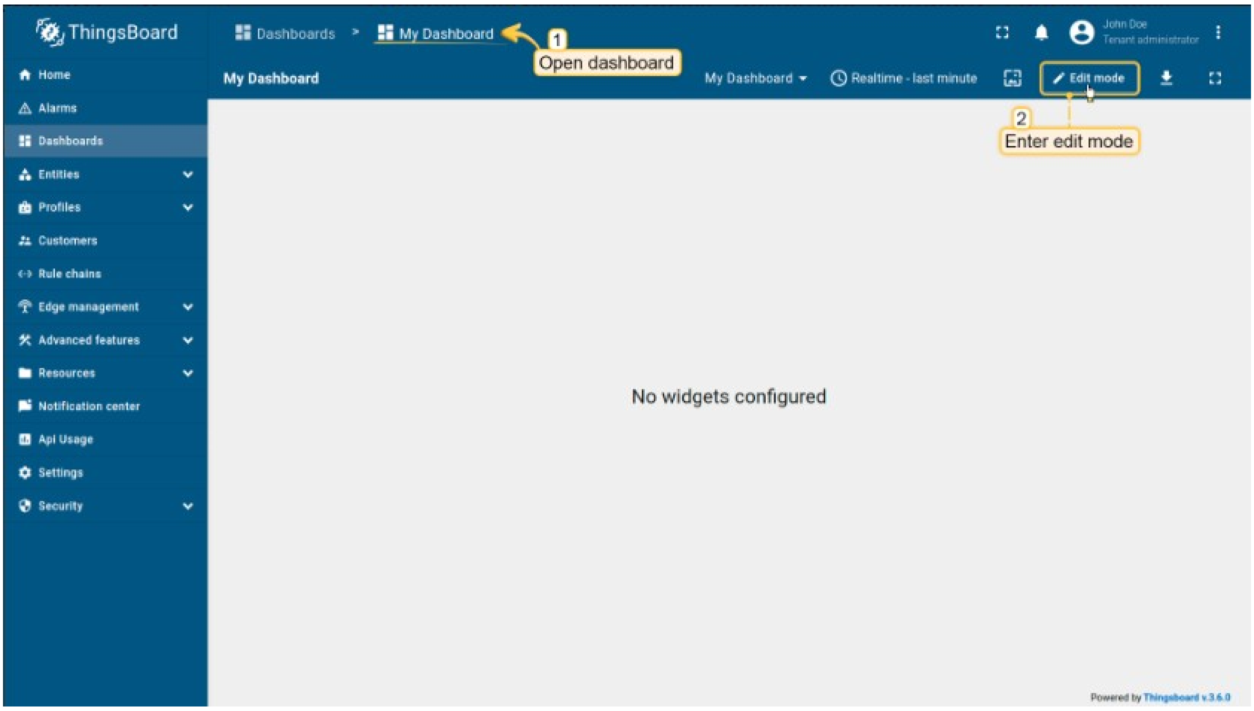
Each widget typically has specific settings and parameters that allow users to customize its behavior and appearance according to their needs.

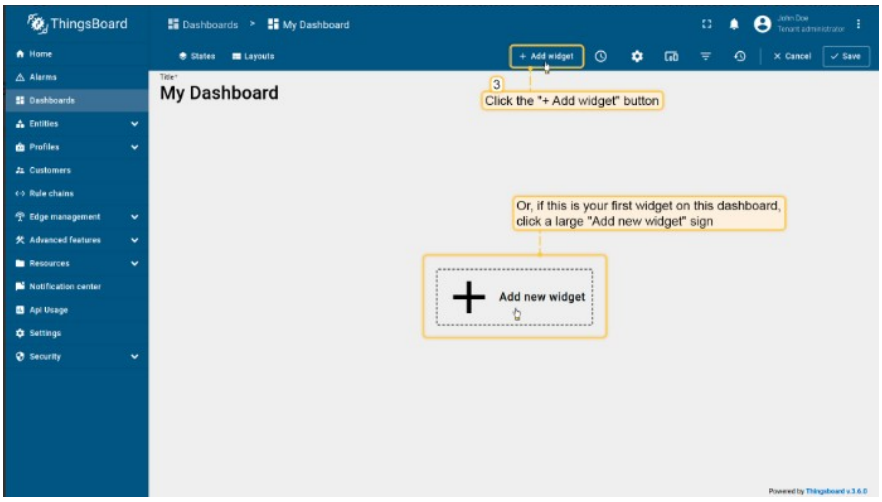
This guide covers main concepts and various widget settings.

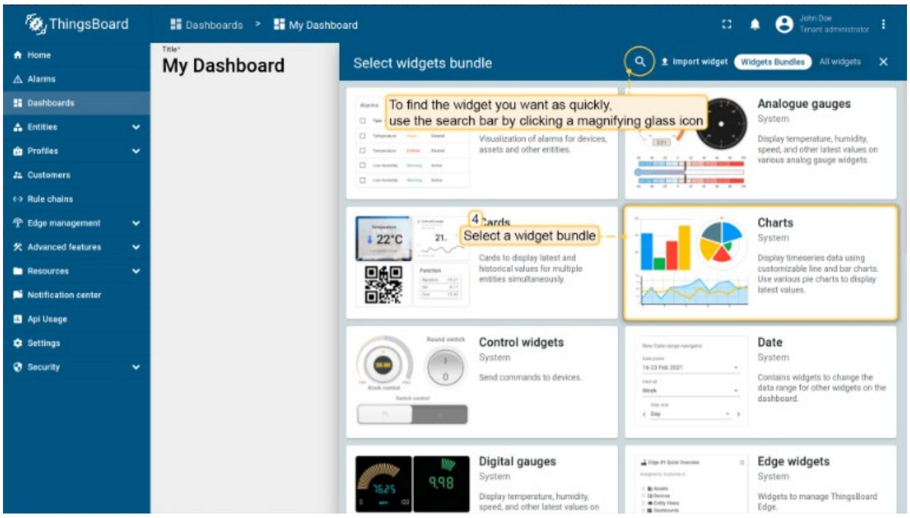
10.9.3 ADDING A WIDGET TO THE DASHBOARD

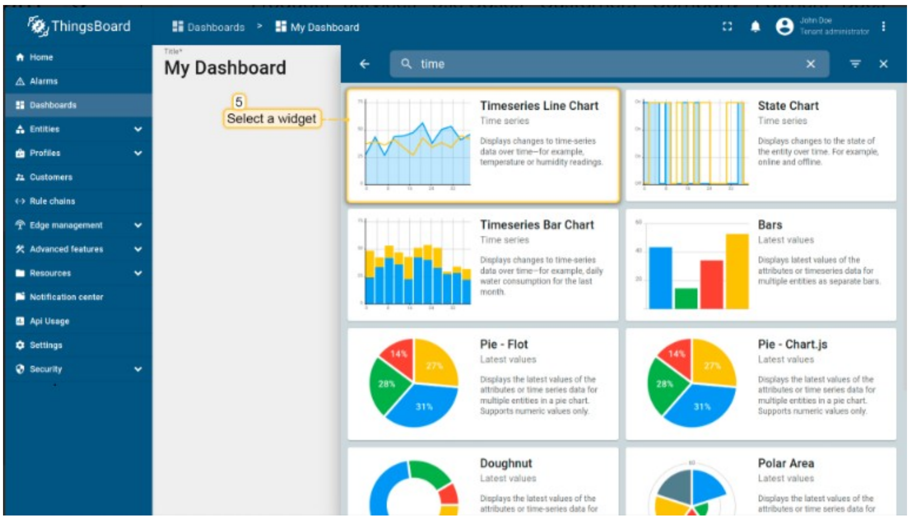
To add a new widget to a dashboard, you should:

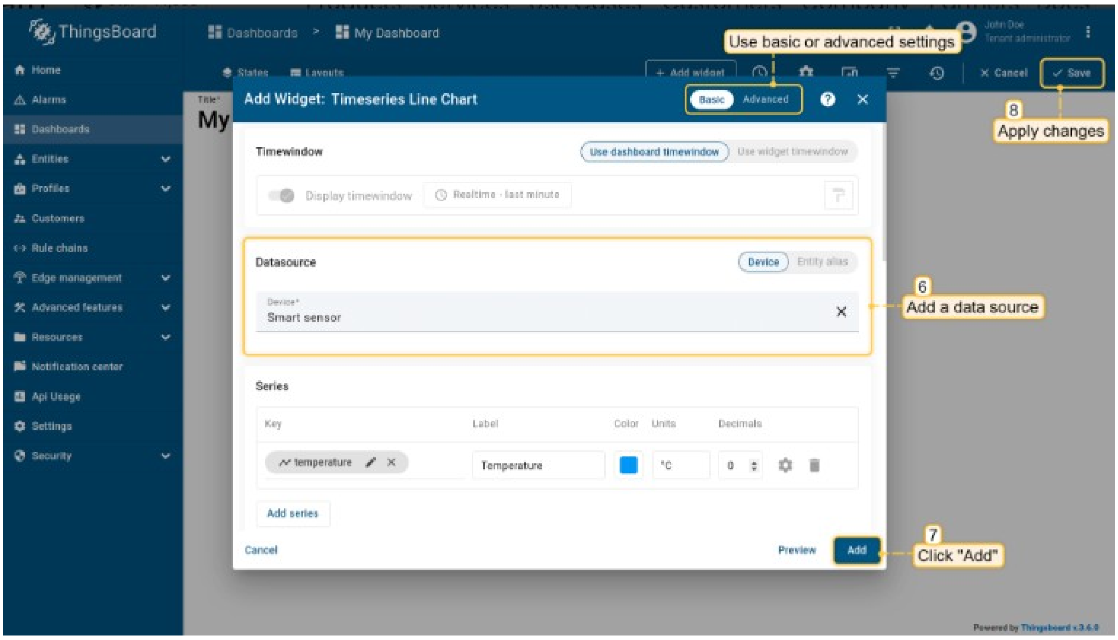
- Open your dashboard and enter edit mode;
- Click the "+ Add widget" icon at the top of the screen, or (if this is your first widget on this dashboard) click a large "Add new widget" sign in the middle of the screen to open the Select widgets bundle dialog window;
- Select a widget bundle, such as "Charts". To quickly find your desired widget, use the search bar by clicking the magnifying glass icon and entering the widget's name;
- Choose a widget, for instance, "Timeseries Line Chart" and click on it to open the "Add Widget" dialog window;
- Specify the data source, add the data key, and click the "Add" button. Finally, apply your changes;
- Your first widget has been created.

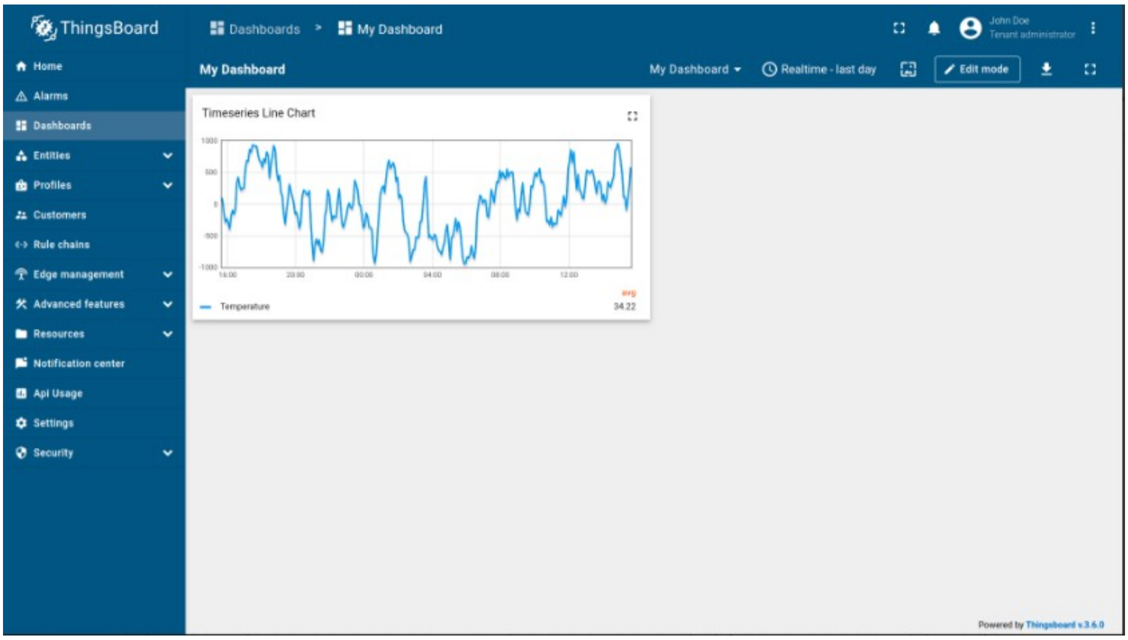



ThingsBoard
Dashboards
My Dashboard
+ Add widget
Cancel
Save
My Dashboard
3
Click the "+ Add widget" button
Or, if this is your first widget on this dashboard,
click a large "Add new widget" sign
Add new widget


ThingsBoard
My Dashboard
Select widgets bundle
Import widget
Widgets Bundles
All widgets
To find the widget you want as quickly,
use the search bar by clicking a magnifying glass icon
Analogue gauges
4
Select a widget bundle
Charts
Control widgets
Date
Digital gauges
Edge widgets


ThingsBoard
My Dashboard
time
5
Select a widget
Timeseries Line Chart
State Chart
Timeseries Bar Chart
Bars
Pie - Flot
Pie - Chart.js
Doughnut
Polar Area


Use basic or advanced settings
8
Apply changes
6
Add a data source
7
Click "Add"
Add Widget: Timeseries Line Chart
Timewindow
Datasource
Series
Add series
Cancel
Preview
Add


ThingsBoard
My Dashboard
Timeseries Line Chart
Temperature

10.10 RASPBERRY PI GPIO CONTROL OVER MQTT USING THINGSBOARD

Introduction

ThingsBoard is an open-source server-side platform that allows you to monitor and control IoT devices. It is free for both personal and commercial usage and you can deploy it anywhere. If this is your first experience with the platform we recommend to review what-is-thingsboard page and getting-started guide.

This sample application will allow you to control GPIO of your Raspberry Pi device using ThingsBoard web UI. We will observe GPIO control using Led connected to one of the pins. The purpose of this application is to demonstrate ThingsBoard RPC capabilities.

Raspberry Pi will use simple application written in Python that will connect to ThingsBoard server via MQTT and listen to RPC commands. Current GPIO state and GPIO control widget is visualized using built-in customizable dashboard.

List of hardware and pinouts

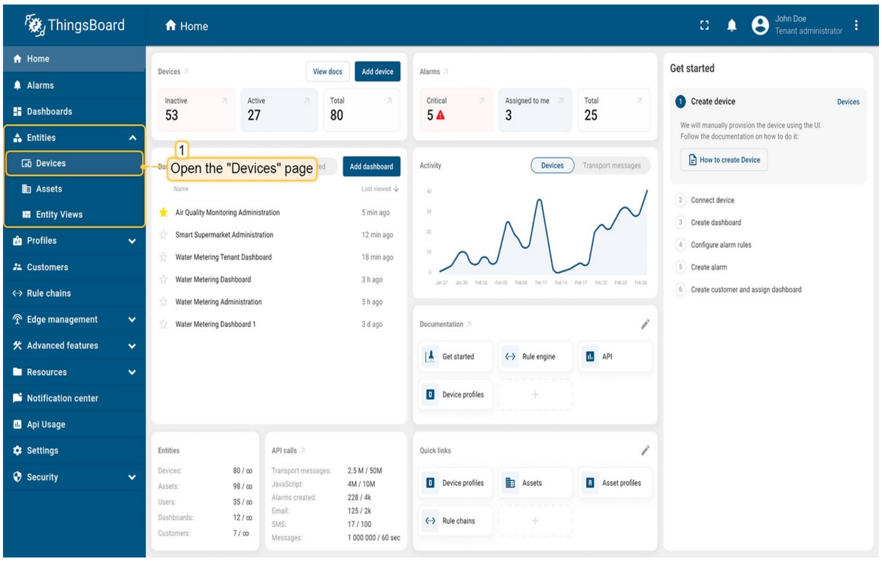
- Raspberry Pi - we will use Raspberry Pi 3 Model B but you can use any other model.
- Led and corresponding resistor

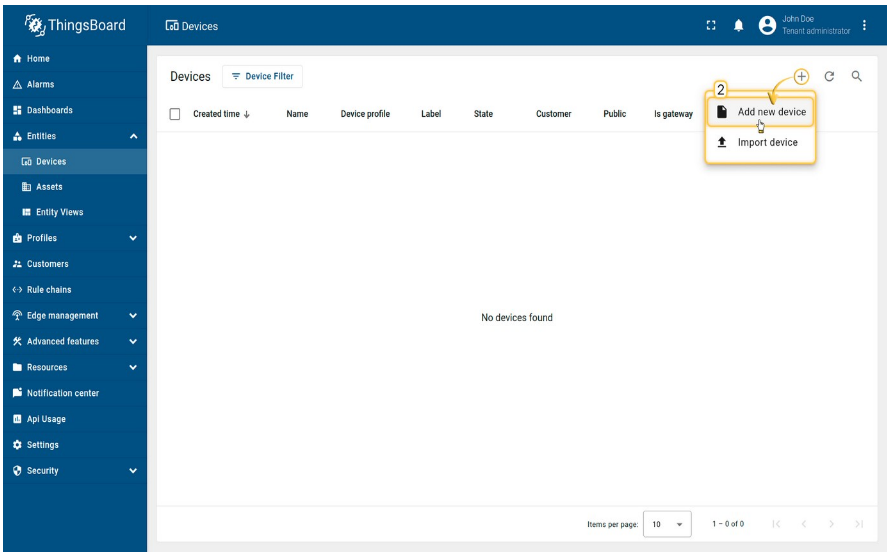
Step 1. Provision Device

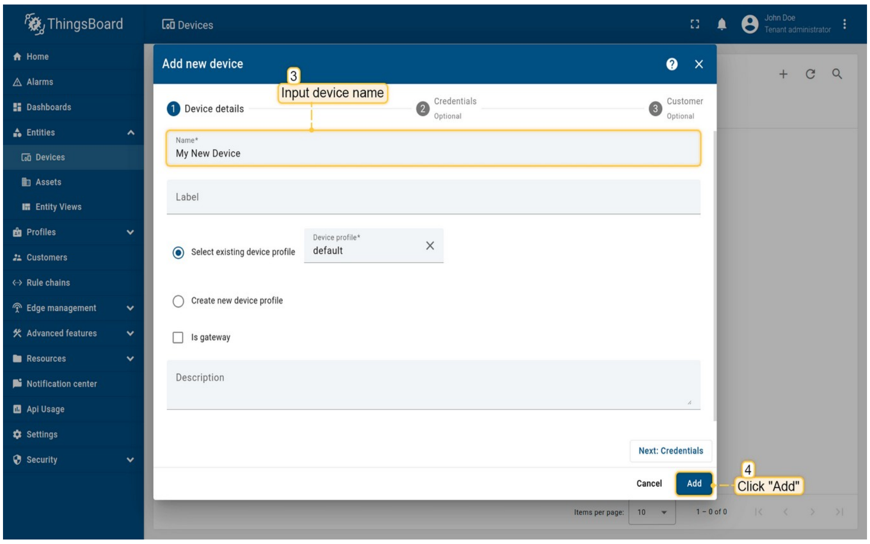
For simplicity, we will provision the device manually using the UI.

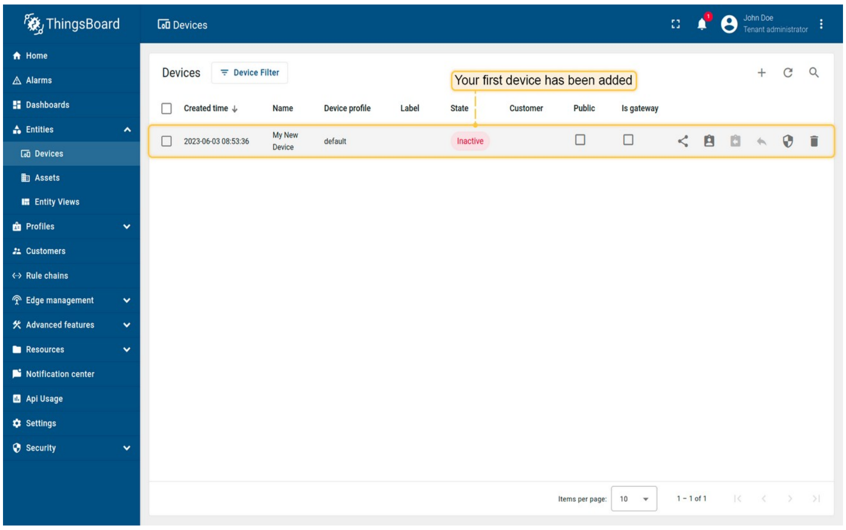
- Login to your ThingsBoard instance and navigate to the "Entities". Then click the "Devices" page;
- Click on the "+" icon in the top right corner of the table and then select "Add new device";

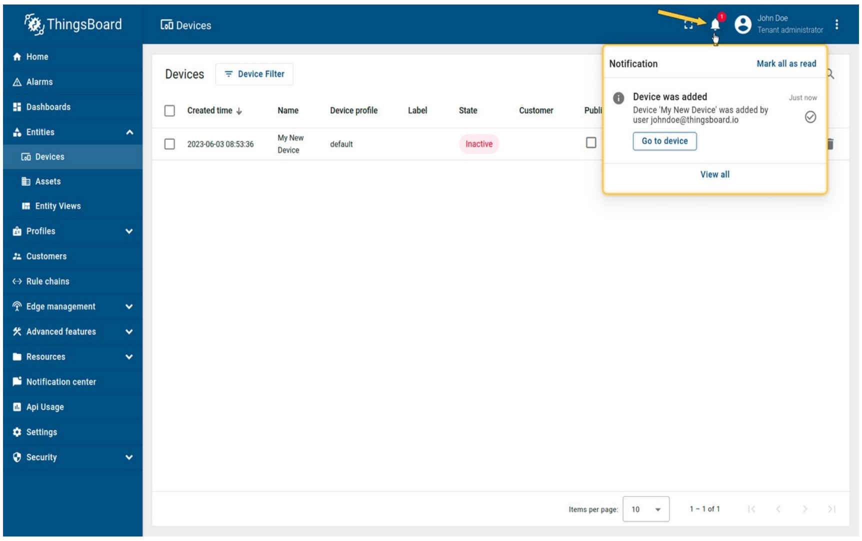
- Input device name. For example, "My New Device". No other changes required at this time. Click "Add" to add the device;
- Your first device has been added. As long as you have one device. But as new devices are added, they will be added to the top of the table, since the table sort devices using the time of the creation by default;
- When adding a new device, you will receive a notification. You can view it by clicking on the "bell" icon in the top right corner.









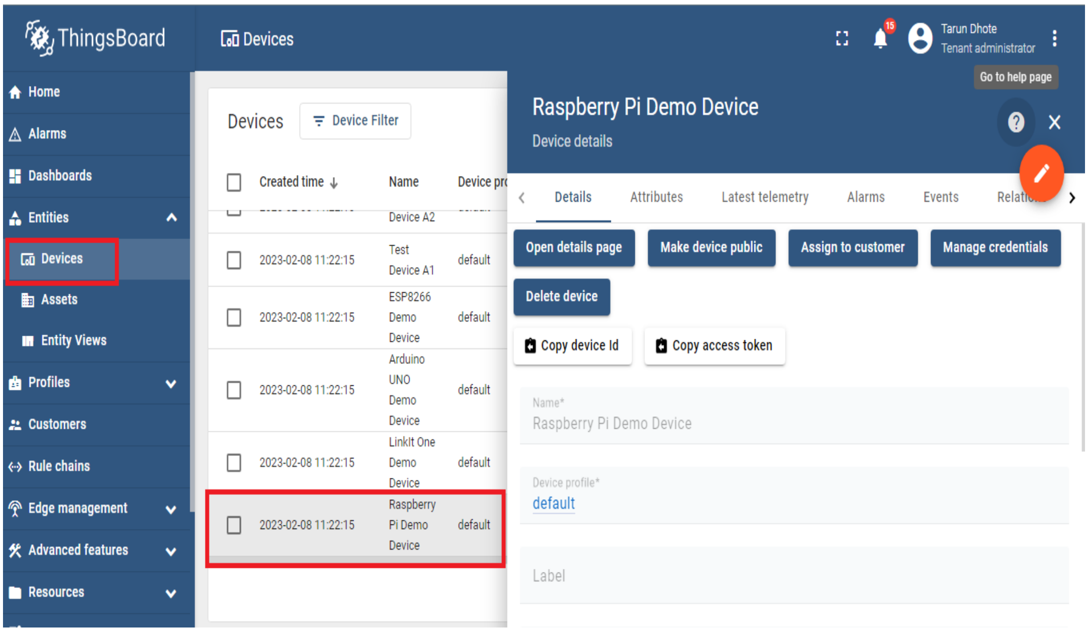


Step 2. Connect device

Go to Devices → Raspberry Pi Demo Device

To connect the device you need to get the device credentials first. ThingsBoard supports various device credentials. We recommend using default auto-generated credentials which is access token for this guide.

- Click on the device row in the table to open device details. Note that the device state is "Inactive";
- Click "Copy access token". Token will be copied to your clipboard. Save it to a safe place.



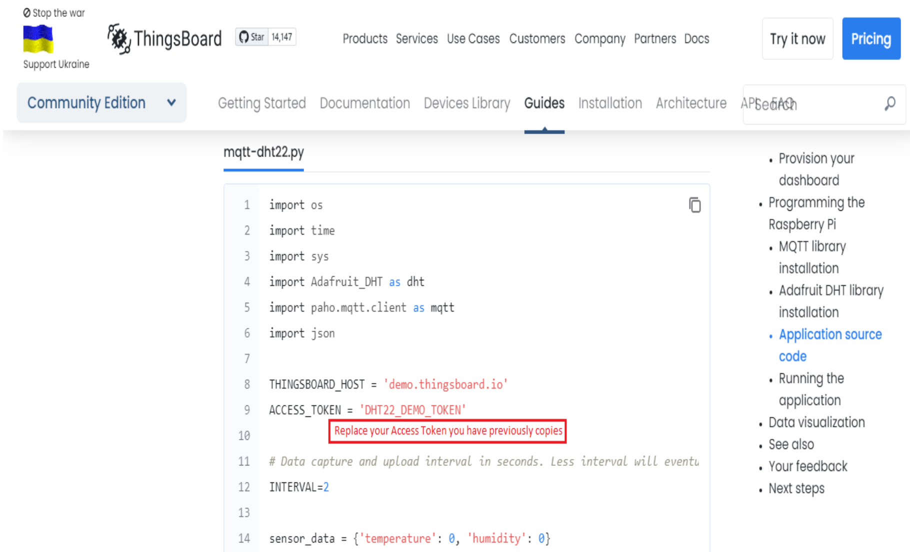
Programming the Raspberry Pi

MQTT library installation

The following command will install MQTT Python library:

- sudo pip install paho-mqtt
- Goto → home/logsun/Documents/Things-board
- Right click on rgb2.py code and open in Thonny IDE

- Replace your access token with your copied access token previously you have done.



- **Click on Run to run the program**

10.11 DATA VISUALIZATION

In order to simplify this guide, we have included "Raspberry PI GPIO Demo Dashboard" to the demo data that is available in each ThingsBoard installation.

in case of local ThingsBoard installation.

Once logged in, open **Dashboards->Raspberry PI GPIO Demo Dashboard** page. You should observe demo dashboard with GPIO control and status panel for your device. Now you can switch status of GPIOs using control panel. As a result, you will see LEDs status change on the device and on the status panel. Below is the screenshot of the “Raspberry PI GPIO Demo Dashboard”.